

RANDOM MATRIX THEORY¹

Unit organiser: Prof. Francesco Mezzadri f.mezzadri@bristol.ac.uk

Lecturers: Prof. Francesco Mezzadri (weeks 13-18) Dr Emma Bailey(weeks 19-24)

Drop-in with lecturers:

- Prof. F. Mezzadri (week 13-19): Monday 16:00-17:00 from 20 January 2025 (week 14) to 24 February 2025 (week 19) in Fry 2A.05
- Emma Bailey (week 20-24): Dr. Emma Bailey. Every Monday between 4pm-5pm from week 20 (3 March 2025) to week 24 (31 March 2025). Office 1A.16

References: There are books (eg. M.L. Mehta, *Random Matrices*, Elsevier, 2004 or P. Forrester *Log-gases and Random Matrices*, Princeton University Press, 2010) on random matrices, and several sets of lecture notes and video lecture series online that you can find by googling, but none follow exactly this course so this set of notes should contain all the background you need. However, in lectures there will be examples, exercises for you to do and discussions that are not included in the notes, so these notes are intended to be read alongside attending lectures.

1 Assessment

- 80% from a 2.5 hour exam in May/June;
- 20% from two homework assignments: **first deadline, Monday 24 February 2025 at 12pm; second deadline, Thursday 3 April 2025 at 12pm.**

Homework is assigned weekly on Friday and will be posted on Blackboard under "Homework and Solutions." The deadline to hand in the homework is the following Thursday at noon. All homework is handed in through a submission point in "Assessment, submission and feedback." For unassessed homework, the submission points are in the subfolder "other formative and draft submissions". Solutions on homework will be posted on Blackboard under "Homework and Solutions" the following week on Friday and any problems/feedback will be discussed in the Friday lecture.

Doing the homework problems (even those which are not assessed!) and handing them in for feedback is the best way to prepare for the exam because this course is not just about calculating, but also about interpreting what you have calculated. So it involves some written answers and you can check that you are able to express your thoughts clearly by getting feedback on the homework.

If you are a graduate student taking this course for credit then you will be assessed on all 10 homeworks — please speak to Francesco and Emma.

The first homework will be assigned on **Friday 14 February 2025**

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2 Applications of Random Matrix Theory

- analytic number theory
- combinatorics
- graph theory
- multivariate statistics
- nuclear physics
- quantum chaos
- quantum information
- statistical mechanics
- structural dynamics
- wireless telecommunications.

3 Why random matrices?

Two reasons:

1. In the limit of large matrix dimension the statistical correlations of the spectra (eigenvalues) of a family, or ensemble, of matrices are independent of the probability distribution that defines the ensemble, but depend only on the invariant properties of such a distribution.
2. RMT techniques allow analytical computations to an extent that is often impossible to achieve in the contexts that they are modelling

One of the aims of the course is to master the techniques in point 2.

One of the first questions students usually ask in this course is “what is an ensemble of matrices?” The rough idea is that you define a set of matrices (such as all $N \times N$ symmetric Hermitian matrices which have real matrix elements) and then define a way to weight matrices in that set (a measure, if you are familiar with that term) so that we can speak of drawing a random matrix from the ensemble. In some ensembles all matrices in the set are equally weighted, and in others the weighting favours matrices with certain properties over others. We will define this more precisely later.

4 Introduction to eigenvalue statistics

4.1 Unitary matrices

A matrix U from the group of all $N \times N$ unitary matrices, $U(N)$, has N eigenvalues lying on the unit circle. We will denote these as $e^{i\theta_1}, e^{i\theta_2}, \dots, e^{i\theta_N}$.

For now let us order the arguments (eigenangles) of the eigenvalues of U on a straight line (although note that many of the calculations we will perform do *not* order the eigenvalues).

$$\theta_1 \leq \theta_2 \leq \dots \leq \theta_N. \tag{1}$$

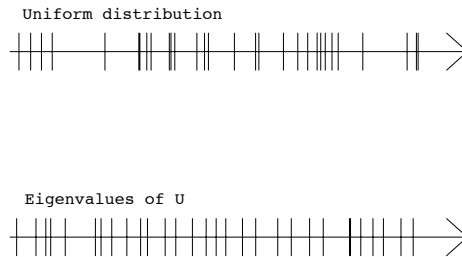


Figure 1: Top figure: points that are each uniformly distributed over the interval and independent of all the others; so, *random* points on an interval. Bottom figure: eigenangles of a random unitary matrix

Do they look like ‘random’ points on the circle? In Figure 1 we can see even by eye that there is a distinct difference between random, uncorrelated points and with eigenangles of a random matrix.

We can write computer code to generate random unitary matrices (with respect to a uniform measure that weights all matrices equally, Haar measure, that we will meet later in the course).

For each $N \times N$ unitary matrix we obtain a list of eigenvalues $e^{i\theta_1}, e^{i\theta_2}, \dots, e^{i\theta_N}$. The eigenangles lie in the range

$$0 < \theta_1 \leq \theta_2 \leq \dots \leq \theta_N \leq 2\pi.$$

They have density $\frac{N}{2\pi}$ (N eigenangles on an interval of length 2π).

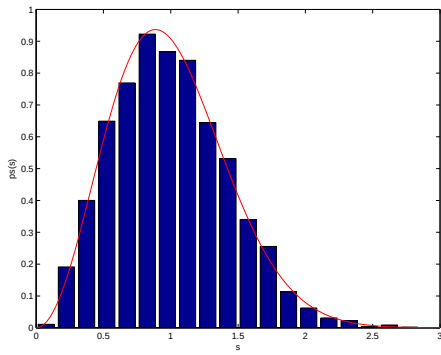
To look at statistics we often scale the eigenangles so that on average the space between consecutive eigenangles is 1:

$$\phi_1 = \theta_1 \times \frac{N}{2\pi}, \dots, \phi_N = \theta_N \times \frac{N}{2\pi}.$$

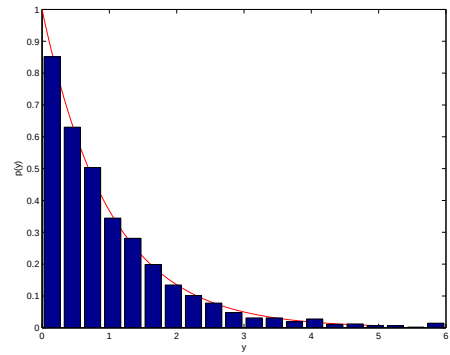
This makes it easier to compare matrices of different sizes.

We compute a list of differences between neighbouring scaled eigenangles: $\phi_2 - \phi_1, \phi_3 - \phi_2, \dots, \phi_N - \phi_{N-1}$. We make a histogram of this list of differences. The histogram can be normalised to have area 1 and so it approximates the probability density of the eigenangle spacings. This is called the “*spacing distribution*” (or sometimes “nearest neighbour spacing distribution”). Integrating the spacing distribution (computing the area under the curve) from 0 to a point a on the x -axis gives the probability that two consecutive eigenangles will have a separation distance between 0 and a .

We can also average this spacing distribution over many random matrices, sampling from an ensemble, see Figures 2 to 7. See Figures 8 and 9 for examples of random matrix statistics occurring in number theory and physics. These figures are for motivation and the reference to various matrix ensembles won’t mean much at the start of the unit, but if you return to them at the end of the unit then you will hopefully understand their significance. If not...ask!

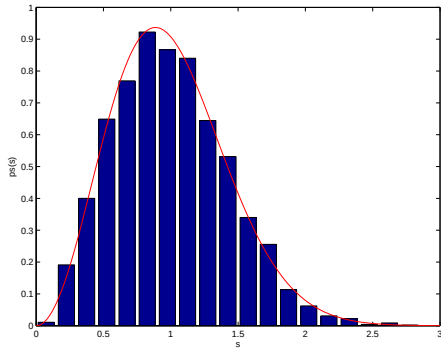


(a) Spacing distribution for eigenangles of one 3000×3000 unitary matrix

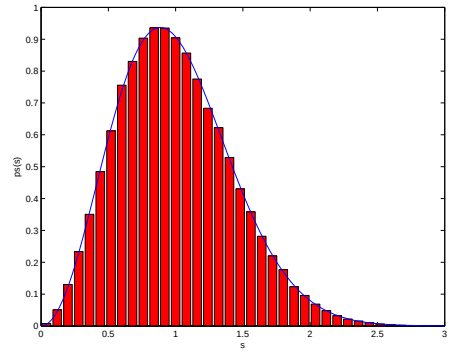


(b) Spacing distribution of 3000 random points uniformly distributed

Figure 2: These plots demonstrate the difference between the spacing distribution of eigenvalues of a random unitary matrix a) as compared to random, uncorrelated points b). The curves are the theoretical distributions that we will meet later in the course.

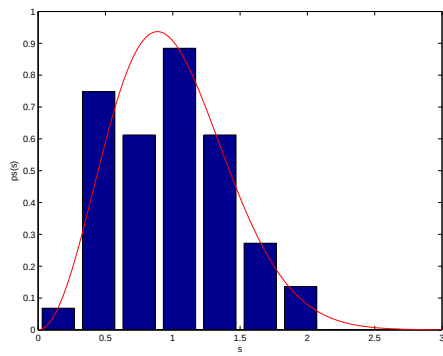


(a) Spacing distribution of one 3000×3000 unitary matrix

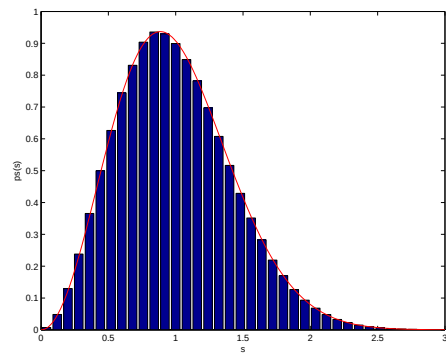


(b) Average spacing distribution for 40 3000×3000 unitary matrices

Figure 3: These plots demonstrate that we have more data and therefore a more smoothly resolved plot if we combine data from many random unitary matrices sampled from the same ensemble. The curves are the theoretical distributions that we will meet later in the course.

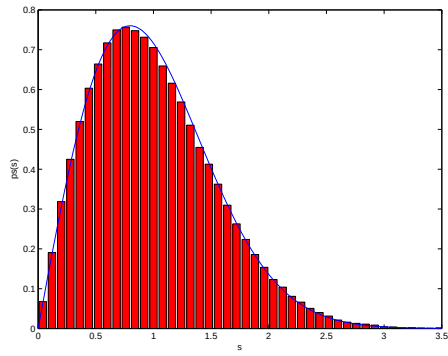


(a) One 50×50 unitary matrix

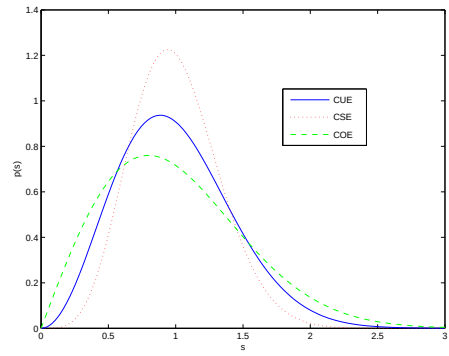


(b) Average over 20000 50×50 unitary matrices

Figure 4: It's even more important to combine data from many matrices and so look at an average over the ensemble when the matrix size is small. The curve is the theoretical distribution (for large matrix size) that we will meet later in the course. We can see it is still a good approximation even for a matrix size of 50.

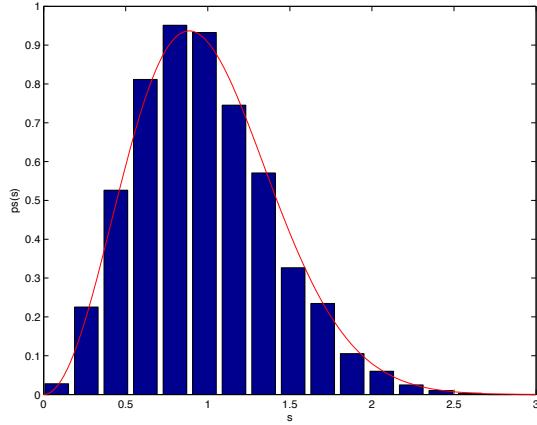


(a) Average over 40 3000×3000 symmetric unitary matrices

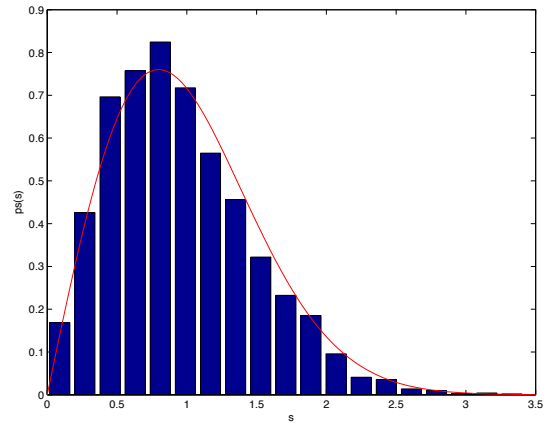


(b) Theoretical curves for the CUE ($U(N)$), COE and CSE ensembles.

Figure 5: Plot a) is of the spacing distribution of symmetric matrices UU^t . Compare with plot b) from Figure 3 to see that the spacing distribution is different if we restrict to *symmetric* unitary matrices. In this course we will meet several different matrix ensembles and plot b) demonstrates that these can have different spacing distributions. These are the theoretical curves for different ensembles



(a) Average over 1000 200×200 Hermitian matrices



(b) Averages over 1000 200×200 real symmetric matrices

Figure 6: We see similar phenomena to the previous figures if we look at Hermitian matrices, with eigenvalues on the real line, rather than unitary matrices. Here the matrix elements are Gaussian random variables.

4.2 Hermitian matrices

We can also look at the spacing distribution between eigenvalues of Hermitian matrices. A matrix $H = (h_{jk})$ is Hermitian if

$$\bar{h}_{jk} = h_{kj}.$$

A 2×2 example of a Hermitian matrix is

$$H = \begin{pmatrix} 3 & 4.5 + i7.5 \\ 4.5 - i7.5 & 6 \end{pmatrix}$$

All the eigenvalues of a Hermitian matrix are real.

5 One dimensional probability distributions.

Suppose we are investigating a quantity X that takes values in $\mathbb{R}$, for example, the time spent queueing at a bus stop, the trace of a random Hermitian matrix, etc. Any well-behaved non-negative real function $p(x)$ normalized to one defines a probability measure on the line. $p(x)$ is called a *probability density function* (*p.d.f.*) of the associated random variable X . We denote the measure of an infinitesimal interval $(x, x + dx)$ by

$$d\mu(x) = p(x)dx.$$

The probability density function is normalised to one if

$$\int_{-\infty}^{\infty} p(x)dx = 1.$$

For any interval $I = (a, b)$, the probability that X lies in I is

$$\mathbb{P}\{X \in I\} = \mu(I) = \int_a^b p(x)dx \geq 0.$$

Probability density functions contain the information to allow us to calculate probabilities.

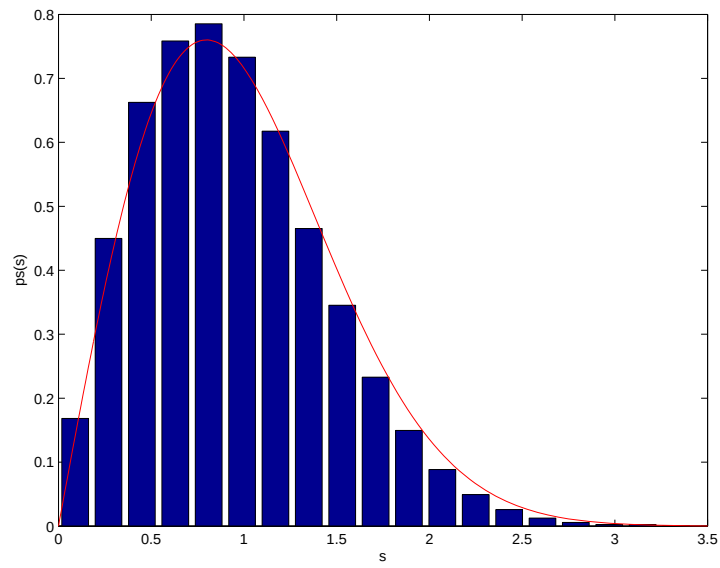


Figure 7: Average over 10000 300×300 symmetric matrices whose *independent elements are either 1 or -1 with 50% probability* compared with the COE curve. Comparing this with Figure 6, plot b), we see that apparently it is the overall symmetry of the matrix that determines the spacing distribution, rather than the details of the distribution on the matrix elements. The theoretical curve on both pictures is the same.

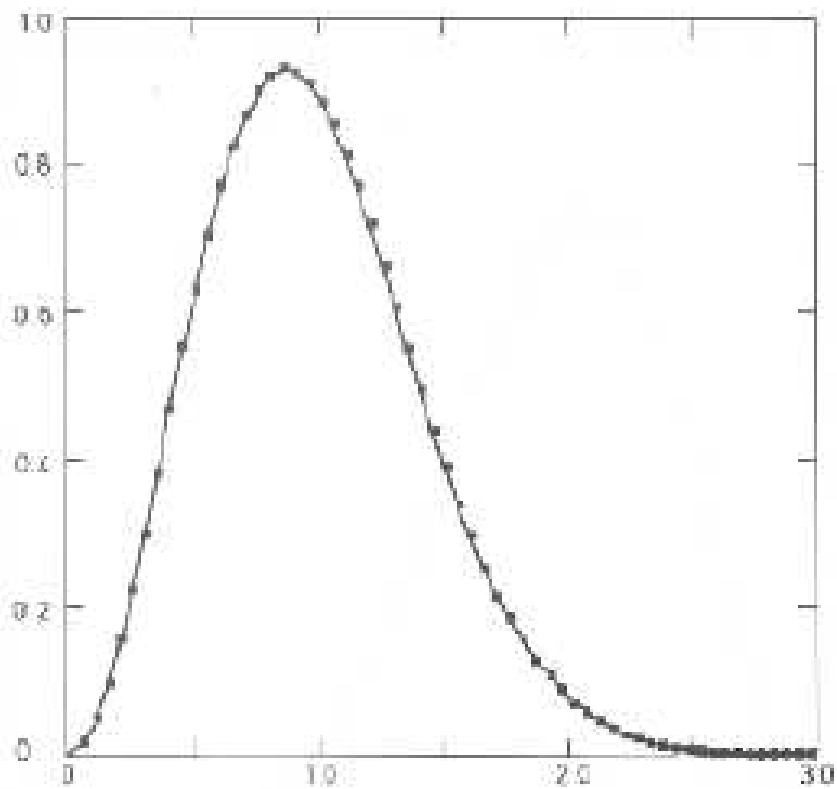


Figure 8: Spacing distribution of the non-trivial zeros of the Riemann zeta function (squares) compared with the CUE ($U(N)$) curve. The data used is 79 millions of zeros around the 10^{20} -th zero. Picture from Odlyzko (1989)

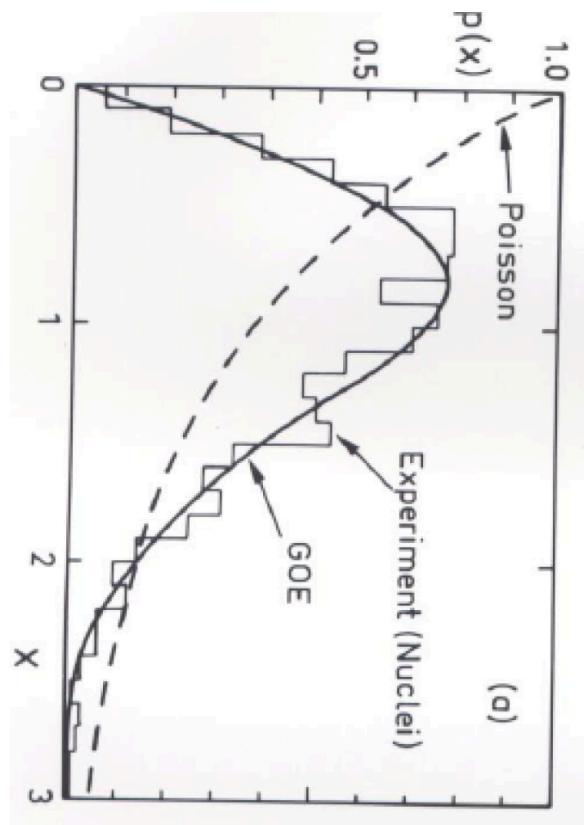


Figure 9: Spacing distribution of the energy levels of heavy atomic nuclei compared with the GOE ensemble of random matrix theory

6 Higher dimensional random variables

If X and Y take values in $\mathbb{R}$ then $\mathbf{Z} = (X, Y)$ takes values in $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$. The probability that $\mathbf{Z}$ lies in A , with $A \subset \mathbb{R}^2$, is

$$\mathbb{P}\{\mathbf{Z} \in A\} = \mu(A) = \int_A p(x, y) dx dy,$$

where $p(x, y)$ is well behaved and non-negative in $\mathbb{R}^2$. $p(x, y)$ is called *joint probability density function* (j.p.d.f.) of X and Y . If $p(x, y) = q(x)r(y)$ we say that the two random variables X and Y are *statistically independent*. In that case, the probability that $X \in I_x = [a, b]$ and $Y \in I_y = [c, d]$ is simply

$$\mu(I_x \times I_y) = \int_a^b \int_c^d q(x)r(y) dx dy = \mu(I_x)\mu(I_y).$$

Not all real random variables live on the full real line. You might have a random variable that lies in $[x_1, x_2]$, in which case you would change $\int_{-\infty}^{\infty}$ to $\int_{x_1}^{x_2}$ in the above definitions.

Remember: it is the PROBABILITIES that need to remain the same when you change to different coordinates, not probability density. Probabilities are the things that can be measured/observed. What is observed can't change depending on what coordinates you use to describe it. So remember the Jacobian when you change variables.

Exercise 1. We have a joint probability density function in $x, y \in \mathbb{R}$:

$$p(x, y) = \frac{1}{\pi} e^{-x^2 - y^2}.$$

We change variables to r and ϕ :

$$r^2 = x^2 + y^2 \quad \text{and} \quad \tan \phi = \frac{y}{x}. \quad (2)$$

Show that the j.p.d.f. in r and ϕ is

$$p(r, \phi) = \frac{1}{\pi} r e^{-r^2}.$$

Solution:

The probability that $(x, y) \in A$, where A is a subset of $\mathbb{R}^2$ is

$$\mathbb{P}\{(x, y) \in A\} = \mu(A) = \frac{1}{\pi} \int_A e^{-x^2 - y^2} dx dy.$$

This probability cannot change if we change coordinates. Thus, all we need to do is to compute the determinant of the Jacobian matrix of the inverse transformation of (2):

$$\begin{aligned} x &= r \cos \phi \\ y &= r \sin \phi \end{aligned}$$

The Jacobian matrix is

$$\frac{\partial(x, y)}{\partial(r, \phi)} = \begin{pmatrix} \cos \phi & -r \sin \phi \\ \sin \phi & r \cos \phi \end{pmatrix}.$$

The absolute value of the determinant of this matrix is

$$|J| = \left| \det \frac{\partial(x, y)}{\partial(r, \phi)} \right| = r.$$

Therefore,

$$\frac{1}{\pi} \int_A e^{-x^2-y^2} dx dy = \frac{1}{\pi} \int_A r e^{-r^2} dr d\phi$$

which is the desired result.

7 Spectral correlation functions

To make the formula look neater, we will define correlation functions here for unitary matrices, however the same ideas would hold for any other ensemble we consider in this unit.

Consider the eigenvalues $e^{i\theta_1}, \dots, e^{i\theta_N}$ of a $N \times N$ unitary matrix U . Take the arguments (note that now they are unordered)

$$0 \leq \theta_1, \dots, \theta_N \leq 2\pi.$$

The *j.p.d.f.* of the $\theta_1, \dots, \theta_N$ for $U(N)$ (the set of all $N \times N$ unitary matrices) with Haar (uniform) measure is

$$P_{N2}(\theta_1, \dots, \theta_N) = \frac{1}{(2\pi)^N N!} \prod_{1 \leq j < k \leq N} \left| e^{i\theta_k} - e^{i\theta_j} \right|^2.$$

Note that this is symmetric in the eigenangles in the sense that if you swapped θ_1 for θ_2 (or swapped any other pair) the formula wouldn't change at all:

$$P_{N2}(\theta_1, \theta_2, \dots, \theta_N) = P_{N2}(\theta_2, \theta_1, \dots, \theta_N).$$

The measure on the infinitesimal interval $(\theta_1, \theta_1 + d\theta_1) \times (\theta_2, \theta_2 + d\theta_2) \times \dots \times (\theta_N, \theta_N + d\theta_N)$ is

$$P_{N2}(\theta_1, \dots, \theta_N) d\theta_1 \dots d\theta_N.$$

In other words, this is the probability of finding eigenangle θ_j in the infinitesimal interval $(\theta_j, \theta_j + d\theta_j)$ for $j = 1, \dots, N$. Note that to make this a *probability* you must include the $d\theta_1 \dots d\theta_N$.

The probability that a random matrix from $U(N)$ (with Haar measure) has eigenangle θ_1 in the interval (a_1, b_1) , θ_2 in the interval (a_2, b_2) and so forth up to θ_N in (a_N, b_N) (note that here the eigenvalues are NOT in ascending order) is

$$\int_{a_1}^{b_1} \int_{a_2}^{b_2} \dots \int_{a_N}^{b_N} P_{N2}(\theta_1, \dots, \theta_N) d\theta_1 \dots d\theta_N.$$

The *n-point correlation function* is defined by

$$R_n(\theta_1, \dots, \theta_n) = \frac{N!}{(N-n)!} \int_0^{2\pi} \cdots \int_0^{2\pi} P_{N2}(\theta_1, \dots, \theta_N) d\theta_{n+1} \cdots d\theta_N$$

For a different ensemble, with eigenvalues $x_1, x_2, \dots, x_N \in [a, b]$ and a *j.p.d.f* of $P(x_1, x_2, \dots, x_N)$, the n -point correlation function is similarly defined as

$$R_n(x_1, \dots, x_n) = \frac{N!}{(N-n)!} \int_a^b \cdots \int_a^b P(x_1, \dots, x_N) dx_{n+1} \cdots dx_N$$

Note that correlation functions are not normalized to one because of the prefactor. Returning to $U(N)$ for concreteness, consider

$$R_1(\theta) = N \int_0^{2\pi} \cdots \int_0^{2\pi} P_{N2}(\theta, \theta_2, \dots, \theta_N) d\theta_2 \cdots d\theta_N$$

$R_1(\theta)$ is called *one-point correlation* or *one-level density*. $R_1(\theta)$ is normalized to N :

$$\int_0^{2\pi} R_1(\theta) d\theta = N.$$

Let $I = [a, b]$ be an interval in $[0, 2\pi)$. The integral

$$\int_a^b R_1(\theta) d\theta$$

gives the number of eigenangles that on average you would expect in I .

Exercise 2 (Solution in lecture). *The unitary symplectic group $USp(2N)$ is a subgroup of $U(2N)$. These are $2N \times 2N$ matrices, A , with $AA^\dagger = 1$ and $A^t J A = J$, where $J = \begin{pmatrix} 0 & I_N \\ -I_N & 0 \end{pmatrix}$ and I_N is the $N \times N$ identity matrix. For these matrices, the eigenvalues lie on the unit circle and come in complex conjugate pairs $e^{i\theta_1}, e^{-i\theta_1}, \dots, e^{i\theta_N}, e^{-i\theta_N}$.*

The joint probability density function on $[0, \pi]^N$ is

$$C_N \prod_{1 \leq j < k \leq N} (\cos \theta_k - \cos \theta_j)^2 \prod_{h=1}^N \sin^2 \theta_h.$$

Here C_N is a normalisation constant.

Consider the integral

$$C_N \int_0^\pi \cdots \int_0^\pi \prod_{1 \leq j < k \leq N} (\cos \theta_k - \cos \theta_j)^2 \prod_{h=1}^N \sin^2 \theta_h d\theta_1 \cdots d\theta_N.$$

We change variables $x_k = \cos \theta_k$, for $k = 1, \dots, N$ in this integral.

a) What is the range of integration of the variables $x_1, \dots, x_N$?

b) Write the integral in terms of the variables $x_1, \dots, x_N$.

There is a well-known ensemble, called the Jacobi ensemble, that depends on two parameters a and b . With the variables x_j in $[-1, 1]$, the *j.p.d.f.* is

$$K \prod_{1 \leq j < k \leq N} (x_k - x_j)^2 \prod_{h=1}^N (1 - x_h)^a (1 + x_h)^b.$$

c) Show that $USp(2N)$ is a case of a Jacobi ensemble. What are the values of a and b in this case?

d) For the case of a Jacobi ensemble with $a = 1$ and $b = 0$ and $N = 2$, calculate the one level density (also called the one-point correlation function).

e) Using part d), for $a = 1$ and $b = 0$ can you say anything about the likelihood of finding an eigenvalue near to 1? Near to -1? Can you read the same behaviour in the j.p.d.f (in the case $N = 2$ or in general)?

f) Does the j.p.d.f tell you anything about whether you are likely to find eigenvalues close together?

Consider the two-point correlation function or pair correlation.

$$R_2(\phi, \chi) = N(N-1) \int_0^{2\pi} \cdots \int_0^{2\pi} P_{N2}(\phi, \chi, \theta_3, \dots, \theta_N) d\theta_3 \cdots d\theta_N.$$

It is normalized so that

$$\int_0^{2\pi} \int_0^{2\pi} R_2(\phi, \chi) d\phi d\chi = N(N-1)$$

is the total number of ordered pairs of eigenangles.

Let $I_1 = [a, b]$ and $I_2 = [c, d]$ be two intervals in $[0, 2\pi)$. The integral

$$\int_a^b \int_c^d R_2(\phi, \chi) d\phi d\chi$$

is the expected number of pairs of eigenangles in $I_1 \times I_2$.

The reason for the prefactor in the definition of the correlation functions is that

$$N(N-1) \cdots (N-n+1) = \frac{N!}{(N-n)!}$$

counts the ways of choosing n ordered θ s from the total N of them.

We have

$$\int_0^{2\pi} \cdots \int_0^{2\pi} R_n(\theta_1, \dots, \theta_n) d\theta_1 \cdots d\theta_n = \frac{N!}{(N-n)!}.$$

Let $D = I_1 \times \cdots \times I_n$ a parallelepiped in $\mathbb{R}^n$. The integral

$$\int_{I_1} \cdots \int_{I_n} R_n(\theta_1, \dots, \theta_n) d\theta_1 \cdots d\theta_n$$

gives the expected number of n -tuples of θ s (regardless of labelling) in D .

Sometimes the integrals involved are hard to perform, so we learn tricks and techniques to tackle them.

Exercise 3 (Solution in lecture). Consider the j.p.d.f. for $U(N)$

$$P_{N2}(\theta_1, \dots, \theta_N) = \frac{1}{(2\pi)^N N!} \prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2.$$

Show that

$$R_1(\theta) = \frac{N}{2\pi}. \quad (3)$$

Hint. You need not perform any integration. Instead, convince yourself that for arbitrary $\alpha \in \mathbb{R}$

$$\begin{aligned} P_{N2}(\theta_1 + \alpha, \dots, \theta_N + \alpha) &= \frac{1}{(2\pi)^N N!} \prod_{1 \leq j < k \leq N} \left| e^{i(\theta_k + \alpha)} - e^{i(\theta_j + \alpha)} \right|^2 \\ &= \frac{1}{(2\pi)^N N!} \prod_{1 \leq j < k \leq N} \left| e^{i\theta_k} - e^{i\theta_j} \right|^2 = P_{N2}(\theta_1, \dots, \theta_N). \end{aligned}$$

Hence deduce (3).

8 Hermitian matrices

$H = (h_{jk})$ is called *Hermitian* or *self-adjoint* if

$$H^\dagger = H \text{ or } h_{jk} = \bar{h}_{kj}.$$

If the elements of H are real, then trivially $H = H^t$ or $h_{jk} = h_{kj}$, i.e. H is *symmetric*.

We will look in depth at one example, the GOE ensemble. Let $X = (x_{jk})$ be an $N \times N$ real symmetric matrix, i.e. $x_{jk} = x_{kj}$. The number of independent parameters is

$$N + \frac{N(N-1)}{2} = \frac{N(N+1)}{2}.$$

Therefore, a symmetric matrix can be identified with a point in $\mathbb{R}^{N(N+1)/2}$.

This is a set of Hermitian matrices. It becomes an ensemble of random matrices when we put a weighting or measure on that set and so can speak of picking a *random* matrix.

Definition 1. *The Gaussian Orthogonal Ensemble (GOE) is the set of all symmetric matrices with matrix elements on and above the diagonal being statistically independent and having the normal distribution with p.d.f.*

$$p(x_{jj}) = \frac{1}{\sqrt{2\pi}} e^{-x_{jj}^2/2} \quad \text{and} \quad p(x_{jk}) = \frac{1}{\sqrt{\pi}} e^{-x_{jk}^2}.$$

This definition puts a probability measure on the space of all symmetric real matrices, or equivalently, $\mathbb{R}^{N(N+1)/2}$. Thus we have made the set of all symmetric matrices into a probability space, called an *ensemble* in RMT.

The *j.p.d.f.* of the matrix elements is

$$\begin{aligned}
P(X) &= \frac{1}{2^{N/2} \pi^{N(N+1)/4}} \prod_{j=1}^N e^{-x_{jj}^2/2} \prod_{1 \leq j < k \leq N} e^{-x_{jk}^2} \\
&= \frac{1}{2^{N/2} \pi^{N(N+1)/4}} e^{-\sum_{j=1}^N x_{jj}^2/2} e^{-\sum_{1 \leq j < k \leq N} x_{jk}^2} \\
&= \frac{1}{2^{N/2} \pi^{N(N+1)/4}} \exp \left(-\frac{1}{2} \left(\sum_{j=1}^N x_{jj}^2 + \sum_{\substack{j,k=1 \\ j \neq k}}^N x_{jk}^2 \right) \right) \\
&= \frac{1}{2^{N/2} \pi^{N(N+1)/4}} \exp \left(-\frac{1}{2} \sum_{j,k=1}^N x_{jk} x_{kj} \right) \\
&= \frac{1}{2^{N/2} \pi^{N(N+1)/4}} \exp \left(-\frac{1}{2} \text{Tr } X^2 \right)
\end{aligned}$$

This representation of the *j.p.d.f.*, although it contains all the statistical information on the matrix X , is not always convenient. We are after the correlations of the spectrum (eigenvalues) of X , how do we extract them from $P(X)$? A symmetric matrix has $N(N+1)/2$ independent parameters. We can choose these in a way that is convenient to us. In other words, we make a convenient change of variables to a set of variables that contains the eigenvalues.

8.1 The 2×2 Gaussian Orthogonal Ensemble

We look at the 2×2 case because we can carry out explicit calculations, but the same steps hold generally for any matrix size. We start out by investigating why normal random variables as the matrix elements is a natural choice.

The set we are dealing with is that of 2×2 symmetric real matrices

$$H = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix},$$

$h_{12} = h_{21}$. Every matrix in this set can be diagonalised by an orthogonal matrix:

$$O^t H O = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

The most general matrix in $O(2)$ is

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The *joint probability density function j.p.d.f.* for the matrix elements is a real non-negative function $P_1(H) = P_1(h_{11}, h_{22}, h_{12})$ such that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P_1(h_{11}, h_{22}, h_{12}) dh_{11} dh_{22} dh_{12} = 1.$$

The probability that the elements of a H lie in a small neighbourhood of h_{11} , h_{22} and h_{12} is

$$P_1(h_{11}, h_{22}, h_{12}) dh_{11} dh_{22} dh_{12}.$$

We want to find the most general $P_1(H) = P_1(h_{11}, h_{22}, h_{12})$ such that

1. If $H' = O^t H O$ for any $O \in O(2)$, we have

$$P_1(h'_{11}, h'_{22}, h'_{12}) dh'_{11} dh'_{22} dh'_{12} = P_1(h_{11}, h_{22}, h_{12}) dh_{11} dh_{22} dh_{12} \quad (4)$$

2. The independent elements of H are uncorrelated, i.e.

$$P_1(h_{11}, h_{22}, h_{12}) = p_{11}(h_{11}) p_{22}(h_{22}) p_{12}(h_{12}).$$

These two conditions can be considered simply as mathematical constraints, but in fact they have real physical significance when modelling, for example, quantum systems in physics: ask the lecturer!

We now take the angle θ to be small, just to make the calculation easier. For small θ $\cos \theta \sim 1 - \frac{\theta^2}{2}$ and $\sin \theta \sim \theta - \frac{\theta^3}{6}$, therefore

$$\delta E = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \sim \begin{pmatrix} 1 & -\theta \\ \theta & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}.$$

This is an approximation to a general matrix from $O(2)$ when the parameter θ is small, but it will serve our purposes.

We then make the change of variables $H' = \delta E^t H \delta E$. Explicitly at first order in θ (that is, we are simply not writing any terms that go to zero faster than θ as $\theta \rightarrow 0$. Terms like θ^2 go to zero faster than θ as $\theta \rightarrow 0$ because you are multiplying something small by something small, so the result is even smaller!)

$$\begin{aligned} h'_{11} &= h_{11} + 2\theta h_{12} \\ h'_{22} &= h_{22} - 2\theta h_{12} \\ h'_{12} &= h_{12} + \theta(h_{22} - h_{11}). \end{aligned} \quad (5)$$

The volume element transforms as

$$dh'_{11} dh'_{22} dh'_{12} = \left| \frac{\partial(h'_{11}, h'_{22}, h'_{12})}{\partial(h_{11}, h_{22}, h_{12})} \right| dh_{11} dh_{22} dh_{12},$$

where

$$\left| \frac{\partial(h'_{11}, h'_{22}, h'_{12})}{\partial(h_{11}, h_{22}, h_{12})} \right| = \begin{vmatrix} 1 & 0 & 2\theta \\ 0 & 1 & -2\theta \\ -\theta & \theta & 1 \end{vmatrix} = 1 + o(\theta)$$

is the *Jacobian*, where $o(\theta)$ represents terms that go to zero faster than θ as $\theta \rightarrow 0$.

Therefore up to, but not including, terms of order θ^2 ,

$$dh'_{11} dh'_{22} dh'_{12} = dh_{11} dh_{22} dh_{12}.$$

This is true even if we had used the exact expression

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

instead of the small θ expansion.

We need to find which functions satisfy

$$\begin{aligned} P_1(H) &= P_1(H') = p_{11}(h_{11}) p_{22}(h_{22}) p_{12}(h_{12}) \\ &= p_{11}(h'_{11}) p_{22}(h'_{22}) p_{12}(h'_{12}) \end{aligned}$$

to satisfy the first condition (4).

By using (5) and expanding at first order in θ we have

$$\begin{aligned}
P_1(H') &= P_1(H) + \theta \left(2h_{12}p'_{11}(h_{11})p_{22}(h_{22})p_{12}(h_{12}) \right. \\
&\quad \left. - 2h_{12}p_{11}(h_{11})p'_{22}(h_{22})p_{12}(h_{12}) \right. \\
&\quad \left. + (h_{22} - h_{11})p_{11}(h_{11})p_{22}(h_{22})p'_{12}(h_{12}) \right) \\
&= P_1(H) \left[1 + \theta \left(2h_{12} \frac{d \log p_{11}}{dh_{11}} - 2h_{12} \frac{d \log p_{22}}{dh_{22}} \right. \right. \\
&\quad \left. \left. + (h_{22} - h_{11}) \frac{d \log p_{12}}{dh_{12}} \right) \right].
\end{aligned}$$

Because θ is arbitrary, the term in the round brackets must be zero. By dividing everything by $h_{12}(h_{11} - h_{22})$ we obtain

$$\frac{1}{h_{12}} \frac{d \log p_{12}}{dh_{12}} - \frac{2}{h_{11} - h_{22}} \left(\frac{d \log p_{11}}{dh_{11}} - \frac{d \log p_{22}}{dh_{22}} \right) = 0.$$

Therefore, we have

$$\begin{aligned}
\frac{1}{h_{12}} \frac{d \log p_{12}}{dh_{12}} &= -4A \\
\frac{1}{h_{11} - h_{22}} \left(\frac{d \log p_{11}}{dh_{11}} - \frac{d \log p_{22}}{dh_{22}} \right) &= -2A.
\end{aligned}$$

Integrating the first equation gives

$$p_{12}(h_{12}) = C \exp(-2Ah_{12}^2).$$

Since $\int_{-\infty}^{\infty} p_{12}(h_{12}) dh_{12} = c$, A must be positive.

The second equation becomes

$$\frac{d \log p_{11}}{dh_{11}} + 2Ah_{11} - \frac{d \log p_{22}}{dh_{22}} - 2Ah_{22} = 0,$$

which gives

$$\begin{aligned}
\frac{d \log p_{11}}{dh_{11}} + 2Ah_{11} &= B \\
\frac{d \log p_{22}}{dh_{22}} + 2Ah_{22} &= B.
\end{aligned}$$

By integrating we obtain

$$\begin{aligned}
p_{11}(h_{11}) &= C_1 \exp(-Ah_{11}^2 + Bh_{11}) \\
\text{and } p_{22}(h_{22}) &= C_2 \exp(-Ah_{22}^2 + Bh_{22}).
\end{aligned}$$

Finally, we obtain

$$\begin{aligned}
P_1(H) &= p_{11}(h_{11})p_{22}(h_{22})p_{12}(h_{12}) \\
&= D \exp(-A(h_{11}^2 + h_{22}^2 + 2h_{12}^2) + B(h_{11} + h_{22})).
\end{aligned} \tag{6}$$

Now we set $A = 1/2$ and $B = 0$ (by convention).

The *j.p.d.f.* for the GOE is therefore

$$P_1(H) = \frac{1}{2\pi^{3/2}} \exp\left(-\frac{1}{2} \text{Tr } H^2\right).$$

Let us denote by $d\mu(H) = dh_{11}dh_{22}dh_{12}$.

Note that the factor in front of the exponential is a normalization constant subject to:

$$\int_{\mathbb{R}^3} P_1(H) d\mu(H) = 1$$

Also, we have obtained the *j.p.d.f.* only by requiring invariance under conjugation by orthogonal matrices and independent elements being uncorrelated.

The *j.p.d.f.* for the matrix elements is a product of *gaussians*, i.e.

$$\begin{aligned}
p_{11}(h_{11}) &= \frac{1}{\sqrt{2\pi}} e^{-h_{11}^2/2} & p_{22}(h_{22}) &= \frac{1}{\sqrt{2\pi}} e^{-h_{22}^2/2} \\
p_{12}(h_{12}) &= \frac{1}{\sqrt{\pi}} e^{-h_{12}^2}
\end{aligned}$$

Remember that the p.d.f. of a Gaussian random variable with mean μ and variance σ^2 is

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-(x - \mu)^2/(2\sigma^2)).$$

8.2 The *j.p.d.f.* for the eigenvalues

Given the pdf

$$P_1(H) = \frac{1}{2\pi^{3/2}} \exp\left(-\frac{1}{2} \text{Tr } H^2\right) \tag{7}$$

we want to know how the eigenvalues λ_1 and λ_2 are distributed.

We want $P(\lambda_1, \lambda_2)$ such that the probability that the eigenvalues λ_1 and λ_2 lie in a small neighbourhood of λ_1 and λ_2 is

$$P_1(\lambda_1, \lambda_2) d\lambda_1 d\lambda_2.$$

In terms of the matrix elements we have three independent variables: h_{11} , h_{22} and h_{12} .

We can diagonalise H :

$$O^t H O = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \text{diag}(\lambda_1, \lambda_2),$$

where

$$O = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

So, a set of independent parameters for H is also λ_1 , λ_2 and θ . We will change variables from h_{11} , h_{22} and h_{12} to λ_1 , λ_2 and θ because then our set of variables contains the eigenvalues that we are interested in. To obtain information about the eigenvalues only, that is $P(\lambda_1, \lambda_2)$, we integrate over θ .

In terms of independent parameters, the formula

$$H = O \operatorname{diag}(\lambda_1, \lambda_2) O^t,$$

with a little work, gives the change of variables

$$h_{11} = \lambda_1 \cos^2 \theta + \lambda_2 \sin^2 \theta \quad (8a)$$

$$h_{22} = \lambda_1 \sin^2 \theta + \lambda_2 \cos^2 \theta \quad (8b)$$

$$h_{12} = \frac{1}{2} \sin 2\theta (\lambda_1 - \lambda_2). \quad (8c)$$

Since a general property of the trace is $\operatorname{Tr} ABA^{-1} = \operatorname{Tr} B$, we have

$$P_1(H) = \frac{1}{2\pi^{3/2}} \exp\left(-\frac{1}{2} \operatorname{Tr} H^2\right) = \frac{1}{2\pi^{3/2}} \exp\left[-\frac{1}{2} (\lambda_1^2 + \lambda_2^2)\right].$$

But when we change variables we need to remember the Jacobian. We want the *probabilities* (integrated probability density) to be independent of the set of variables we choose to work with. The Jacobian matrix for this transformation is

$$\begin{aligned} J(\lambda_1, \lambda_2, \theta) &= \frac{\partial(h_{11}, h_{22}, h_{12})}{\partial(\lambda_1, \lambda_2, \theta)} \\ &= \begin{pmatrix} \cos^2 \theta & \sin^2 \theta & \sin 2\theta(\lambda_2 - \lambda_1) \\ \sin^2 \theta & \cos^2 \theta & -\sin 2\theta(\lambda_2 - \lambda_1) \\ \frac{1}{2} \sin 2\theta & -\frac{1}{2} \sin 2\theta & -\cos 2\theta(\lambda_2 - \lambda_1) \end{pmatrix}. \end{aligned}$$

We have

$$\begin{aligned} |\det J(\lambda_1, \lambda_2, \theta)| &= |\lambda_2 - \lambda_1| \times \begin{vmatrix} \cos^2 \theta & \sin^2 \theta & \sin 2\theta \\ \sin^2 \theta & \cos^2 \theta & -\sin 2\theta \\ \frac{1}{2} \sin 2\theta & -\frac{1}{2} \sin 2\theta & -\cos 2\theta \end{vmatrix} \\ &= |\lambda_2 - \lambda_1|. \end{aligned}$$

Note that the Jacobian $\det J(\lambda_1, \lambda_2, \theta)$ does not depend on θ .

We have

$$\begin{aligned} \frac{1}{2\pi^{3/2}} \exp\left[-\frac{1}{2} (h_{11}^2 + h_{22}^2 + 2h_{12}^2)\right] dh_{11} dh_{22} dh_{12} \\ = \frac{1}{2\pi^{3/2}} |\lambda_2 - \lambda_1| \exp\left[-\frac{1}{2} (\lambda_1^2 + \lambda_2^2)\right] d\lambda_1 d\lambda_2 d\theta. \end{aligned}$$

Integrating over θ in $[0, \pi/2)$ (can you explain why?), we obtain

$$P_1(\lambda_1, \lambda_2) = \frac{1}{4\sqrt{\pi}} |\lambda_2 - \lambda_1| \exp\left[-\frac{1}{2} (\lambda_1^2 + \lambda_2^2)\right].$$

This procedure generalizes to the GOE of matrices of dimension $N \times N$, but is just more complicated to work through. An $N \times N$ symmetric matrix has $N(N+1)/2$ independent elements. In short, the Jacobian of the transformation from the h_{jk} ($j \leq k$) to the eigenvalues λ_j ($j = 1, \dots, N$) and $N(N-1)/2$ real variables p_k that parametrize the degrees of freedom of the eigenvectors factorizes into the product

$$\left(\prod_{1 \leq j < k \leq N} |\lambda_k - \lambda_j| \right) f(p_1, \dots, p_{N(N-1)/2}).$$

This allows us to integrate easily over the p_j s to obtain

$$P_1(\lambda_1, \dots, \lambda_N) = C \prod_{1 \leq j < k \leq N} |\lambda_k - \lambda_j| \exp \left(-\frac{1}{2} \sum_{j=1}^N \lambda_j^2 \right). \quad (9)$$

This is because

$$\exp \left(-\frac{1}{2} \text{Tr } X^2 \right) = \exp \left(-\frac{1}{2} \sum_{j=1}^N \lambda_j^2 \right). \quad (10)$$

Note that the exponential part of the joint probability density function in (9) depends on the particular measure that we have chosen, that is on the way the matrix elements are generated, in this case as Gaussian variables. However, the factor $\prod_{1 \leq j < k \leq N} |\lambda_k - \lambda_j|$ comes from the Jacobian, and therefore depends only on the symmetries, or geometry, of the ensemble.

The product

$$\Delta_N(x_1, \dots, x_N) := \prod_{1 \leq j < k \leq N} (x_k - x_j) = \det V \quad (11)$$

is a *Vandermonde determinant*. That is, it is the determinant of the Vandermonde matrix:

$$V = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{N-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{N-1} \\ 1 & x_3 & x_3^2 & \cdots & x_3^{N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_N & x_N^2 & \cdots & x_N^{N-1} \end{pmatrix}.$$

Equivalently we can write $V = (x_j^{k-1})$, where j parametrises the rows and k the columns. The product $\prod_{1 \leq j < k \leq N} |\lambda_k - \lambda_j|$ is therefore the absolute value of a Vandermonde determinant.

Exercise 4 (Solution in lecture). *The normalized j.p.d.f. for the eigenvalues of the GOE of 2×2 matrices is given by*

$$P(\lambda_1, \lambda_2) = \frac{1}{4\sqrt{\pi}} |\lambda_2 - \lambda_1| e^{-\frac{1}{2}(\lambda_1^2 + \lambda_2^2)}.$$

Compute the one-level density

$$R_1(\lambda) = 2 \int_{-\infty}^{\infty} P(\lambda_1, \lambda) d\lambda_1.$$

Express the result using the error function

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt. \quad (12)$$

You may need the integral

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}.$$

Definition 2. The Gaussian Unitary Ensemble (GUE) is the set of all complex $N \times N$ Hermitian (self-adjoint) matrices with the following joint probability density function for the matrix elements:

$$\begin{aligned} P_2(H) &= \frac{2^{N(N-1)/2}}{\pi^{N^2/2}} \prod_{j=1}^N \exp(-h_{jj}^2) \prod_{1 \leq j < k \leq N} \exp(-2|h_{jk}|^2) \\ &= \frac{2^{N(N-1)/2}}{\pi^{N^2/2}} \exp(-\operatorname{Tr} H^2). \end{aligned}$$

We could carry out all of the calculations that we have done for the GOE in Section 8 in a similar manner for the 2×2 GUE. We would find the *j.p.d.f.* of the eigenvalues is

$$P_2(\lambda_1, \lambda_2) = \frac{1}{\pi} (\lambda_1 - \lambda_2)^2 \exp(-\lambda_1^2 - \lambda_2^2). \quad (13)$$

The square on the Vandermonde factor is a result of the larger number of degrees of freedom in the GUE with its complex matrix elements, and it means there is quadratic repulsion between eigenvalues rather than the linear repulsion in the GOE. $P_2(\lambda_1, \lambda_2)$ is normalized to one, that is

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P_2(\lambda_1, \lambda_2) d\lambda_1 d\lambda_2 = 1. \quad (14)$$

Exercise 5. Using the *j.p.d.f.* for the 2×2 GUE in (13), compute the one-level density

$$R_1(\lambda) = 2 \int_{-\infty}^{\infty} P_2(x, \lambda) dx. \quad (15)$$

Solution: We have

$$\begin{aligned} R_1(\lambda) &= \frac{2}{\pi} \int_{-\infty}^{\infty} (\lambda - x)^2 e^{-\lambda^2 - x^2} dx = \frac{2}{\pi} \int_{-\infty}^{\infty} (\lambda^2 - 2\lambda x + x^2) e^{-\lambda^2 - x^2} dx \\ &= \frac{2}{\pi} \int_{-\infty}^{\infty} (\lambda^2 + x^2) e^{-\lambda^2 - x^2} dx = \frac{2\lambda^2 e^{-\lambda^2}}{\pi} \int_{-\infty}^{\infty} e^{-x^2} dx + \frac{2e^{-\lambda^2}}{\pi} \int_{-\infty}^{\infty} x^2 e^{-x^2} dx \\ &= \frac{2e^{-\lambda^2}}{\sqrt{\pi}} \left(\lambda^2 + \frac{1}{2} \right), \end{aligned} \quad (16)$$

where we have used the fact that xe^{-x^2} is an odd function and therefore its integral is zero. The integral $\int_{-\infty}^{\infty} x^2 e^{-x^2} dx$ can be computed using integration by parts:

$$\int_{-\infty}^{\infty} x^2 e^{-x^2} dx = -\frac{xe^{-x^2}}{2} \Big|_{-\infty}^{\infty} + \frac{1}{2} \int_{-\infty}^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}. \quad (17)$$

9 Spacing distribution of a Poisson sequence

Consider N independent identically distributed, *i.i.d.*, random points in the interval $[0, L]$:

$$0 \leq x_1, x_2, \dots, x_N \leq L.$$

The probability measure of each x_j is the uniform distribution:

$$p(x_j) = \frac{1}{L} = \rho.$$

The *j.p.d.f.* of $x_1, \dots, x_N$ is

$$p(x_1, \dots, x_N) = \frac{1}{L^N}.$$

and the one-level density is

$$R_1(x) = \frac{N}{L^N} \int_0^L \dots \int_0^L dx_2 \dots dx_N = \frac{N}{L}.$$

The number of expected points in an interval $I = [a, b]$ of length $l = b - a$ is

$$\frac{N}{L} \int_a^b dx = \frac{Nl}{L}.$$

We can generalize this to an infinite sequence of random points. Let

$$\dots x_j < x_{j+1} < x_{j+2} < \dots < x_{j+N} \dots$$

have the following properties:

1. The probability that the interval $(x, x+dx)$ contains a point is ρdx and is independent of x , *i.e. it is constant.*
2. The above probability does not depend on the positions of the other points, *i.e. they are uncorrelated.*

Now we consider the sequence of the distances between adjacent points: $s_j = x_{j+1} - x_j$. We will call the s_j s *spacings*. The question we want to answer is, what is the probability that a spacing s should have a certain value? That is, what is the conditional probability that:

1. Given a point at x ;
2. the interval $(x, x+s)$ is empty;
3. there is a point in the interval $(x+s, x+s+ds)$?

To work this out, we work through the following steps. First we divide the interval s in n parts, each of length s/n . An interval of length s/n is either empty or contains one or more point. If s/n is small enough we can neglect the probability that it contains more than one point.

The probability that s/n contains a point is $\rho(s/n)$; the probability that it is empty is $1 - \rho(s/n)$. Therefore, the probability that $(x, x + s)$ is empty is given by the probability that each of the n parts is empty, i.e.

$$\left(1 - \frac{\rho s}{n}\right)^n.$$

By taking the limit as $n \rightarrow \infty$ we obtain

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\rho s}{n}\right)^n = e^{-\rho s}.$$

This is the probability that the interval $(x, x + s)$ is empty.

The probability that there is a point in $(x + s, x + s + ds)$ is ρds .

So, combining these, the probability that given a point at x the next point is at $(x + s, x + s + ds)$ is

$$p(s)ds = \rho e^{-\rho s} ds.$$

Note that we have

$$\int_0^\infty p(s)ds = \rho \int_0^\infty e^{-\rho s} ds = 1.$$

If we rescale the coordinates $x_j \mapsto \rho x_j$, so that the mean spacing for the new sequence is one, the probability density function p.d.f. for the level spacings is $p(s) = e^{-s}$.

10 The Wigner surmise

In the previous section we saw that the probability density of the spacings in a Poisson sequence (with mean spacing normalized to one) is $p(s) = e^{-s}$. Now we want to compute the spacing distribution of the eigenvalues of the 2×2 GOE.

For a 2×2 matrix the spacing between consecutive eigenvalues is simply $s = |\lambda_2 - \lambda_1|$. Consider the *j.p.d.f.* for the eigenvalues

$$P_1(\lambda_1, \lambda_2) = \frac{1}{4\sqrt{\pi}} |\lambda_2 - \lambda_1| \exp \left[-\frac{1}{2} (\lambda_1^2 + \lambda_2^2) \right].$$

We change to a more convenient set of variables:

$$\mu_1 = \lambda_1 - \lambda_2 \quad \text{and} \quad \mu_2 = \lambda_1 + \lambda_2.$$

We have

$$\left| \frac{\partial(\lambda_1, \lambda_2)}{\partial(\mu_1, \mu_2)} \right| = \frac{1}{2}.$$

Furthermore,

$$s = |\lambda_2 - \lambda_1| = |\mu_1| \quad \text{and} \quad \lambda_1^2 + \lambda_2^2 = \frac{1}{2} (s^2 + \mu_2^2)$$

Substituting into $P(\lambda_1, \lambda_2)$ we have

$$P_1(s, \mu_2) = \frac{s}{8\sqrt{\pi}} \exp \left[-\frac{1}{4} (s^2 + \mu_2^2) \right],$$

then integrating over μ_2 we obtain

$$p_1(s) = C s \exp(-\alpha s^2).$$

Here the constants C and α are determined by the two conditions

$$\int_0^\infty p_1(s) ds = 1 \quad \text{and} \quad \int_0^\infty s p_1(s) ds = 1.$$

The first condition just says that the total probability of finding a spacing of any length must be one; the second condition just fixes the mean spacing to be one.

First condition:

$$C \int_0^\infty s \exp(-\alpha s^2) ds = 1, \quad \frac{C}{2\alpha} = 1.$$

Second condition:

$$2\alpha \int_0^\infty s^2 \exp(-\alpha s^2) ds = 1, \quad \frac{1}{2} \sqrt{\frac{\pi}{\alpha}} = 1.$$

Finally, we obtain the *Wigner surmise*:

$$p_1(s) = \frac{\pi s}{2} \exp\left(-\frac{\pi}{4} s^2\right).$$

An important point is that the Wigner surmise is exact only for the GOE of 2×2 matrices, but gives a very good approximation to the spacing distribution of matrices in the GOE of arbitrary large dimensions.

We could instead have done this exercise for 2×2 complex Hermitian matrices (dropping the condition of symmetry). As we have seen at Definition 2, the Gaussian Unitary Ensemble (GUE) of 2×2 matrices is the space of complex Hermitian 2×2 matrices endowed with the following joint probability density function for the matrix elements:

$$P_2(H) = \frac{2}{\pi^2} \exp(-\text{Tr } H^2) = \frac{2}{\pi^2} \exp\left(-h_{11}^2 - h_{22}^2 - 2|h_{12}|^2\right).$$

The *j.p.d.f.* for the eigenvalues is

$$P_2(\lambda_1, \lambda_2) = \frac{1}{\pi} |\lambda_2 - \lambda_1|^2 \exp(-\lambda_1^2 - \lambda_2^2).$$

The spacing distribution is

$$p_2(s) = \frac{32s^2}{\pi^2} \exp\left(-\frac{4s^2}{\pi}\right).$$

As with the GOE, this spacing distribution is exact for 2×2 matrices and a very good approximation for higher dimensions.

It is important to note that there is a difference in the exponent of $|\lambda_2 - \lambda_1|$ between the GOE and GUE ensembles. This determines the strength of the level repulsion. We can see this difference in repulsion near zero on the x -axis of the plots in Figure 6.

11 Summary of matrix ensembles we have (and have not!) met so far

- *Gaussian ensembles*: Gaussian Unitary Ensemble (GUE), Gaussian Orthogonal ensemble (GOE) (GSE, the Gaussian Symplectic Ensemble, we haven't met - uses quaternions).

These are ensembles of Hermitian matrices, where the matrix elements are the appropriate (complex, real, quaternion) normal random variables.

These are often used in applications to physics and engineering.

- *Classical compact groups*: Unitary group $U(N)$, Orthogonal group $O(N)$, Unitary symplectic group $USp(2N)$

These are ensembles (groups) of unitary matrices with Haar measure (a “flat” measure weighting all matrices equally) on the group. $U(N)$ is the set of all $N \times N$ unitary matrices. $O(N)$ is the set of all $N \times N$ orthogonal matrices. Orthogonal matrices with determinant 1 and those with determinant -1 have different eigenvalue statistics and are often considered separately. We write, for example, $SO(2N)$ for $2N \times 2N$ orthogonal matrices with determinant 1. $USp(2N)$ is the set of all $2N \times 2N$ matrices that are also symplectic (see Exercise 2 for the definition).

These are often used in applications to number theory.

- *Circular ensembles*: The most familiar of these is the CUE (Circular unitary ensemble), which is identical to $U(N)$ with Haar measure - physicists tend to say CUE and pure mathematicians $U(N)$. We haven't formally encountered the COE (Circular orthogonal ensemble) and CSE (Circular symplectic ensemble). These three circular ensembles are the unitary analogue of the GUE, GOE and GSE, in the sense that they are invariant under transformations by unitary, orthogonal or symplectic matrices respectively, and the eigenvalue statistics are the same as the Hermitian version in the large matrix limit. Note that while the CUE is the same as $U(N)$, the COE and CSE are NOT equal to $O(N)$ and $USp(2N)$.
- *Poisson ensemble*: These are diagonal matrices with independent entries. The eigenvalues are independent random points.
- $\beta = 2$ ensembles In Section 14 we will meet a collection of ensembles which all have a *j.p.d.f.* of the specific form given in (48). Here the “ $\beta = 2$ ” refers to the exponent on the Vandermonde factor in the *j.p.d.f.*. We will learn some techniques with orthogonal polynomials to deal with these ensembles. The Jacobi ensemble is one example of this which already popped up in Exercise 2.

12 Eigenvalue statistics for the unitary group $U(N)$, also called the CUE ensemble

We are going to spend some time working through detailed calculations for the CUE. Similar calculations can be carried out for other random matrix ensembles.

12.1 The joint probability density function of the eigenvalues

The starting point to study statistics of eigenvalues of random matrices is the *joint probability density function (j.p.d.f.) for the eigenvalues*. This is a real positive function that we will denote by

$$P_{N2}(\theta_1, \theta_2, \dots, \theta_N),$$

with the condition

$$\int_0^{2\pi} \int_0^{2\pi} \cdots \int_0^{2\pi} P_{N2}(\theta_1, \theta_2, \dots, \theta_N) d\theta_1 d\theta_2 \cdots d\theta_N = 1.$$

The probability that a matrix taken at random will have the arguments of its eigenvalues lying in the intervals

$$\theta_1^i < \theta_1 < \theta_1^f, \quad \theta_2^i < \theta_2 < \theta_2^f, \quad \dots, \quad \theta_N^i < \theta_N < \theta_N^f$$

is given by

$$\int_{\theta_1^i}^{\theta_1^f} \int_{\theta_2^i}^{\theta_2^f} \cdots \int_{\theta_N^i}^{\theta_N^f} P_{N2}(\theta_1, \theta_2, \dots, \theta_N) d\theta_1 d\theta_2 \cdots d\theta_N.$$

For the CUE ensemble we state, without proof, the formula for the joint probability density function of eigenvalues:

$$P_{N2}(\theta_1, \dots, \theta_N) = \frac{1}{(2\pi)^N N!} \prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2.$$

12.2 Correlation functions revisited

In the cases that we will consider the *j.p.d.f.* P_N , will be invariant under permutations of its arguments:

$$P_N(\theta_1, \theta_2, \dots, \theta_N) = P_N(\theta_{\sigma 1}, \theta_{\sigma 2}, \dots, \theta_{\sigma N}),$$

where σ is any permutation of $[1, N] = \{1, \dots, N\}$. We then define the n -th correlation function as

$$R_n(\theta_1, \dots, \theta_n) = \frac{N!}{(N-n)!} \int_0^{2\pi} \cdots \int_0^{2\pi} P_N(\theta_1, \dots, \theta_N) d\theta_{n+1} \cdots d\theta_N.$$

Remember that $R_1(\theta)d\theta$ gives the expected number of arguments of the complex numbers $e^{i\theta_1}, \dots, e^{i\theta_N}$ in a small neighbourhood of θ .

For the two-point correlation function, there are $N(N-1)$ ways of choosing an ordered pair of θ s among N : $R_2(\theta_1, \theta_2)d\theta_1 d\theta_2$ gives the expected number of such pairs where the first is in a small neighbourhood of θ_1 and the second in a small neighbourhood of θ_2 .

There are $N(N-1) \cdots (N-n+1) = N!/(N-n)!$ ways of choosing n ordered θ s among N . $R_n(\theta, \dots, \theta_n)d\theta_1 \cdots d\theta_n$ gives the expected number of such n -tuples that lie in a small neighbourhood of $\theta_1, \dots, \theta_n$.

It is interesting to note that the spacing distribution requires knowledge of all R_n s - that is, it incorporates information about n -tuples of eigenvalues for all values of n .

Our main goal will be to compute

$$R_n^{\text{U}(N)}(\theta_1, \dots, \theta_n) = \frac{N!}{(N-n)!} \int_0^{2\pi} \cdots \int_0^{2\pi} P_{N2}(\theta_1, \dots, \theta_N) d\theta_{n+1} \cdots d\theta_N.$$

We can also write this as

$$R_n^{\text{U}(N)}(\theta_1, \dots, \theta_n) = \frac{N!}{(N-n)!} \int_{[0, 2\pi]^{(N-n)}} P_{N2}(\theta_1, \dots, \theta_N) d\theta_{n+1} \cdots d\theta_N.$$

12.3 Where we are headed

Many random matrix ensembles have a beautiful structure that we can exploit to calculate eigenvalue statistics. In particular, the *j.p.d.f* and correlation functions can often be written as determinants. We will explore this structure for the CUE.

To start we need some matrix notation. Firstly, if $f(x, y)$ is a function of two variables $(f(x_j, x_k))$ denotes the square matrix

$$(f(x_j, x_k)) = \begin{pmatrix} f(x_1, x_1) & \cdots & f(x_1, x_n) \\ \vdots & \ddots & \vdots \\ f(x_n, x_1) & \cdots & f(x_n, x_n) \end{pmatrix}. \quad (18)$$

Unless specified it could be of any dimensions.

If we add indices, then with $(f(x_j, x_k))_{lm}$ we denote the element on the l -th row and m -th column of the matrix (18).

We often need to denote determinants of matrices of different sizes, so

$$\det_{n \times n}(f(x_j, x_k))$$

denotes the determinant of the matrix (18) of dimension $n \times n$.

And now we can state the result we are aiming at:

1. To begin with we will prove that

$$\prod_{1 \leq j < k \leq N} \left| e^{i\theta_k} - e^{i\theta_j} \right|^2 = (2\pi)^N \det_{N \times N}(S_N(\theta_k - \theta_j)), \quad (19)$$

where

$$S_N(\theta) = \frac{1}{2\pi} \frac{\sin\left(\frac{N\theta}{2}\right)}{\sin\frac{\theta}{2}}.$$

2. Then we will show that

$$R_n^{\text{U}(N)}(\theta_1, \dots, \theta_n) = \det_{n \times n}(S_N(\theta_k - \theta_j)). \quad (20)$$

A *very important* point is that $R_n^{\text{U}(N)}$ depends only on the difference of the arguments θ_j . This is a consequence of the fact that the l.h.s. of Eq. (19) is invariant under the mapping $\theta_j \mapsto \theta_j + \alpha$.

12.4 Correlation functions for the CUE ensemble ($U(N)$)

From (20) (which we will accept for now - the proof will come later)

$$R_1^{U(N)}(\theta) = \det_{1 \times 1} (S_N(\theta_k - \theta_j)).$$

Of course, a 1×1 matrix ($S_N(\theta_k - \theta_j)$) is simply

$$S_N(\theta_1 - \theta_1) = S_N(0) = \lim_{\theta \rightarrow 0} \frac{1}{2\pi} \frac{\sin\left(\frac{N\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)} = \frac{N}{2\pi}.$$

Therefore $R_1^{U(N)}(\theta) = \frac{N}{2\pi}$.

For the two point correlation function (pair correlation), when $n = 2$ the matrix ($S_N(\theta_k - \theta_j)$) is

$$\frac{1}{2\pi} \begin{pmatrix} N & \frac{\sin\left(\frac{N(\theta_1 - \theta_2)}{2}\right)}{\sin\left(\frac{\theta_1 - \theta_2}{2}\right)} \\ \frac{\sin\left(\frac{N(\theta_2 - \theta_1)}{2}\right)}{\sin\left(\frac{\theta_2 - \theta_1}{2}\right)} & N \end{pmatrix}.$$

Therefore, we have

$$\begin{aligned} R_2^{U(N)}(\theta_1, \theta_2) &= \det_{2 \times 2} (S_N(\theta_k - \theta_j)) \\ &= \frac{N^2}{(2\pi)^2} \left[1 - \left(\frac{\sin\left(\frac{N(\theta_2 - \theta_1)}{2}\right)}{N \sin\left(\frac{\theta_2 - \theta_1}{2}\right)} \right)^2 \right]. \end{aligned}$$

12.5 Correlation functions in the large N limit

Sometimes we want to compare $R_2^{U(N)}(\theta_1, \theta_2)$ (in general $R_n^{U(N)}$) for different N . To do this we need to scale the argument of the eigenvalues so that their density is equal to 1 for each N :

$$x_j = \frac{N\theta_j}{2\pi} \tag{21a}$$

$$\tilde{R}_1^{U(N)}(x) = \frac{2\pi}{N} R_1^{U(N)}\left(\frac{2\pi x}{N}\right) = 1 \tag{21b}$$

$$\tilde{R}_2^{U(N)}(x_1, x_2) = \left(\frac{2\pi}{N}\right)^2 R_2^{U(N)}\left(\frac{2\pi x_1}{N}, \frac{2\pi x_2}{N}\right) \tag{21c}$$

$$\tilde{R}_n^{U(N)}(x_1, \dots, x_n) = \left(\frac{2\pi}{N}\right)^n R_n^{U(N)}\left(\frac{2\pi x_1}{N}, \dots, \frac{2\pi x_n}{N}\right). \tag{21d}$$

The factor $(2\pi/N)^n$ in front of $R_n^{U(N)}$ comes from the normalization constraint

$$\int_0^N \dots \int_0^N \tilde{R}_n^{U(N)}(x_1, \dots, x_n) dx_1 \dots dx_n = \frac{N!}{(N-n)!} \tag{22}$$

and from

$$d\theta_1 \dots d\theta_n = \left(\frac{2\pi}{N}\right)^n dx_1 \dots dx_n.$$

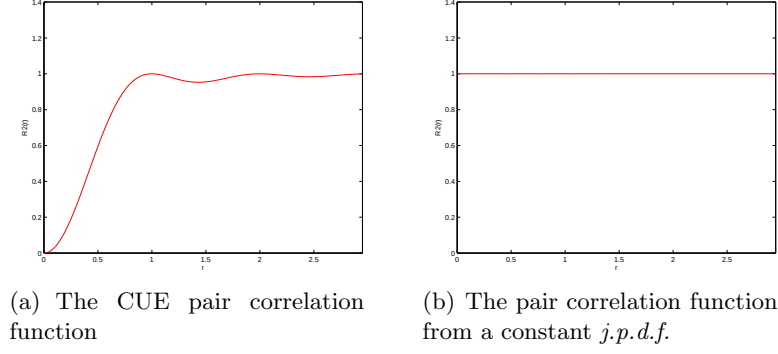


Figure 10: Comparison between the CUE pair correlation function and that one of a set of random numbers uniformly distributed.

This scaling of the eigenvalues makes the average difference between consecutive eigenvalues (spacing) equal to 1 for each N . In this way we can compare *local* correlations as $N \rightarrow \infty$.

When we take the limit $N \rightarrow \infty$ it is clear that the limit of the integral (22) doesn't exist. However, we will see that the limit

$$R_n(x_1, \dots, x_n) = \lim_{N \rightarrow \infty} \tilde{R}_n^{U(N)}(x_1, \dots, x_n) \quad (23)$$

does exist.

We have

$$R_2^{U(N)}(\theta_1, \theta_2) = \frac{N^2}{(2\pi)^2} \left[1 - \left(\frac{\sin\left(\frac{N(\theta_2 - \theta_1)}{2}\right)}{N \sin\left(\frac{\theta_2 - \theta_1}{2}\right)} \right)^2 \right].$$

By setting $r = N(\theta_2 - \theta_1)/(2\pi)$ we have

$$\begin{aligned} \frac{\sin\left(\frac{N(\theta_2 - \theta_1)}{2}\right)}{N \sin\left(\frac{\theta_2 - \theta_1}{2}\right)} &= \frac{\sin(\pi r)}{N \sin\left(\frac{\pi r}{N}\right)} = \frac{\sin(\pi r)}{\pi r - \frac{1}{N^2} \frac{(\pi r)^3}{3!} + O\left(\frac{1}{N^4}\right)} \\ &= \frac{\sin(\pi r)}{\pi r} \left(1 + \frac{1}{N^2} \frac{(\pi r)^2}{3!} + O\left(\frac{1}{N^4}\right) \right). \end{aligned}$$

Here $O\left(\frac{1}{N^4}\right)$ indicates all terms that go to zero as fast as or faster than a constant times $1/N^4$, eg $1/N^4$ terms, $1/N^5$ terms, etc.

Therefore, we have

$$\left(\frac{2\pi}{N}\right)^2 R_2^{U(N)}\left(\frac{2\pi r}{N}\right) = 1 - \left(\frac{\sin(\pi r)}{\pi r}\right)^2 + O\left(\frac{1}{N^2}\right). \quad (24)$$

Finally, by taking the limit $N \rightarrow \infty$ we obtain

$$R_2(r) = 1 - \left(\frac{\sin(\pi r)}{\pi r}\right)^2.$$

Note that Eq. (24) implies that for numerical purposes the convergence to $R_2(r)$ is fairly fast. Averaging over a high enough number of matrices, for small r and $N = 100$ the discrepancy with $R_2(r)$ should be of the order of 10^{-4} .

By expanding $\sin(\pi r)/(\pi r)$ around zero we obtain

$$\frac{\sin(\pi r)}{\pi r} = \frac{\pi r - \frac{(\pi r)^3}{3!} + O((\pi r)^5)}{\pi r} = 1 - \frac{(\pi r)^2}{3!} + O((\pi r)^4).$$

Here $O((\pi r)^5)$ means all terms in the expansion of $\sin$ that go to zero as fast as or faster than a constant times r^5 , so powers of r with exponent 5 or higher.

Therefore,

$$\left(\frac{\sin(\pi r)}{\pi r} \right)^2 \sim 1 - \frac{(\pi r)^2}{3}, \quad r \rightarrow 0,$$

$$R_2(r) \sim \frac{(\pi r)^2}{3}, \quad r \rightarrow 0.$$

This quadratic behaviour for small r is a consequence of the exponent 2 in

$$\prod_{1 \leq j < k \leq N} \left| e^{i\theta_k} - e^{i\theta_j} \right|^2.$$

This behaviour can be seen at the origin of Figure 10 a).

Similarly, we can look in the $N \rightarrow \infty$ limit at the n -point correlation. The limit as $N \rightarrow \infty$ of

$$\left(\frac{2\pi}{N} \right)^n R_n^{\text{U}(N)} \left(\frac{2\pi x_1}{N}, \dots, \frac{2\pi x_n}{N} \right)$$

can easily be taken in the same way as the limit for the pair correlation. We have that

$$S_N(\theta) = \frac{1}{2\pi} \frac{\sin\left(\frac{N\theta}{2}\right)}{\sin\frac{\theta}{2}}.$$

Therefore, by setting $r_{kj} = N(\theta_k - \theta_j)/(2\pi)$ we obtain

$$\begin{aligned} (S_N(\theta_k - \theta_j)) &= (S_N(2\pi r_{kj}/N)) \\ &= \frac{N}{2\pi} \begin{pmatrix} \frac{\sin(\pi r_{11})}{N \sin \frac{\pi r_{11}}{N}} & \cdots & \frac{\sin(\pi r_{1n})}{N \sin \frac{\pi r_{1n}}{N}} \\ \vdots & \ddots & \vdots \\ \frac{\sin(\pi r_{n1})}{N \sin \frac{\pi r_{n1}}{N}} & \cdots & \frac{\sin(\pi r_{nn})}{N \sin \frac{\pi r_{nn}}{N}} \end{pmatrix}. \end{aligned}$$

We then define

$$S(r) = \lim_{N \rightarrow \infty} \frac{\sin(\pi r)}{N \sin \frac{\pi r}{N}} = \frac{\sin(\pi r)}{\pi r}.$$

Finally, we obtain

$$\begin{aligned} R_n(x_1, \dots, x_n) &= \lim_{N \rightarrow \infty} \left(\frac{2\pi}{N} \right)^n R_n^{\text{U}(N)} \left(\frac{2\pi x_1}{N}, \dots, \frac{2\pi x_n}{N} \right) \\ &= \lim_{N \rightarrow \infty} \left(\frac{2\pi}{N} \right)^n \det_{n \times n} S_N(2\pi r_{kj}/N) \\ &= \det_{n \times n} (S(r_{kj})). \end{aligned}$$

12.6 Haar measure as a determinant

The first step to compute the n -point correlations as a determinant is to show

$$\prod_{1 \leq j < k \leq N} \left| e^{i\theta_k} - e^{i\theta_j} \right|^2 = (2\pi)^N \det_{N \times N} (S_N(\theta_k - \theta_j)). \quad (25)$$

The second step is to prove

$$R_n^{U(N)}(\theta_1, \dots, \theta_n) = \det_{n \times n} (S_N(\theta_k - \theta_j)).$$

Consider the matrix $(\psi_j(x_k))$:

$$\begin{pmatrix} \psi_1(x_1) & \cdots & \psi_1(x_n) \\ \vdots & \ddots & \vdots \\ \psi_n(x_1) & \cdots & \psi_n(x_n) \end{pmatrix}. \quad (26)$$

The transpose of Vandermonde matrix,

$$V^t = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ x_1 & x_2 & x_3 & \cdots & x_n \\ x_1^2 & x_2^2 & x_3^2 & \cdots & x_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & x_3^{n-1} & \cdots & x_n^{n-1} \end{pmatrix},$$

is of this type with $\psi_j(x_k) = x_k^{j-1}$.

Lemma (Transposing lemma). *We have*

$$\det_{n \times n} (\psi_j(x_k)) \det_{n \times n} (\phi_j(x_k)) = \det_{n \times n} \left(\sum_{l=1}^n \psi_l(x_j) \phi_l(x_k) \right).$$

Exercise 6. *Prove the transposing lemma.*

Use the fact that if A and B are two square matrices, then

$$\det A \det B = \det A^t \det B = \det A^t B$$

Then write matrix multiplication as sum over matrix elements.

Proof. We have

$$\begin{aligned} \det_{n \times n} (\psi_m(x_p)) \det_{n \times n} (\phi_m(x_p)) &= \det_{n \times n} (\psi_m(x_p))^t \det_{n \times n} (\phi_m(x_p)) \\ &= \det_{n \times n} (\psi_m(x_p))^t (\phi_m(x_p)). \end{aligned}$$

Furthermore,

$$\begin{aligned} ((\psi_m(x_p))^t (\phi_m(x_p)))_{jk} &= \sum_{l=1}^n (\psi_m(x_p))^t_{jl} (\phi_m(x_p))_{lk} \\ &= \sum_{l=1}^n (\psi_m(x_p))_{lj} (\phi_m(x_p))_{lk} = \sum_{l=1}^n \psi_l(x_j) \phi_l(x_k). \end{aligned}$$

Combining these two equations we obtain

$$\det_{n \times n}(\psi_j(x_k)) \det_{n \times n}(\phi_j(x_k)) = \det_{n \times n} \left(\sum_{l=1}^n \psi_l(x_j) \phi_l(x_k) \right).$$

□

We have

$$\begin{aligned} \prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2 &= \Delta(e^{i\theta_1}, \dots, e^{i\theta_N}) \Delta(e^{-i\theta_1}, \dots, e^{-i\theta_N}) \\ &= \det_{N \times N} \left(e^{i(j-1)\theta_k} \right) \det_{N \times N} \left(e^{-i(j-1)\theta_k} \right). \end{aligned} \quad (27)$$

We set $\psi_j(\theta_k) = e^{i(j-1)\theta_k}$ and $\phi_j(\theta_k) = e^{-i(j-1)\theta_k}$ and apply the transposing lemma:

$$\prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2 = \det_{N \times N} \left(\sum_{l=1}^N e^{i(l-1)(\theta_k - \theta_j)} \right). \quad (28)$$

The matrix elements in the r.h.s. are geometric sums.

We then apply the formula

$$\sum_{k=0}^{N-1} x^k = \frac{1 - x^N}{1 - x}$$

and obtain

$$\sum_{l=1}^N e^{i(l-1)(\theta_k - \theta_j)} = \sum_{l=0}^{N-1} e^{il(\theta_k - \theta_j)} = \begin{cases} \frac{1 - e^{iN(\theta_k - \theta_j)}}{1 - e^{i(\theta_k - \theta_j)}} & \text{if } \theta_k \neq \theta_j \\ N & \text{if } \theta_k = \theta_j. \end{cases}$$

Then, we can write

$$\begin{aligned} \frac{1 - e^{iN(\theta_k - \theta_j)}}{1 - e^{i(\theta_k - \theta_j)}} &= e^{i\frac{N-1}{2}(\theta_k - \theta_j)} \frac{e^{iN(\theta_k - \theta_j)/2} - e^{-iN(\theta_k - \theta_j)/2}}{e^{i(\theta_k - \theta_j)/2} - e^{-i(\theta_k - \theta_j)/2}} \\ &= e^{i\frac{N-1}{2}(\theta_k - \theta_j)} \frac{\sin(N(\theta_k - \theta_j)/2)}{\sin((\theta_k - \theta_j)/2)} \\ &= 2\pi e^{i\frac{(N-1)\theta_k}{2}} S_N(\theta_k - \theta_j) e^{-i\frac{(N-1)\theta_j}{2}}. \end{aligned} \quad (29)$$

Combining the previous formula with Eq. (28) we obtain

$$\begin{aligned} &\prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2 \\ &= \det_{N \times N} \left(2\pi e^{i(N-1)\theta_k/2} S_N(\theta_k - \theta_j) e^{-i(N-1)\theta_j/2} \right) \\ &= (2\pi)^N \prod_{n=1}^N e^{i(N-1)\theta_n/2} \prod_{m=1}^N e^{-i(N-1)\theta_m/2} \\ &\quad \times \det_{N \times N} (S_N(\theta_k - \theta_j)) \\ &= (2\pi)^N \det_{N \times N} (S_N(\theta_k - \theta_j)), \end{aligned}$$

Exercise 7. We can follow a similar procedure for other ensembles and write the j.p.d.f as a determinant. Let's take just the first step at (27). The j.p.d.f. (ignoring the normalisation constant) of $USp(2N)$ (see Exercise 2 for the full definition) is the square of

$$\prod_{1 \leq j < k \leq N} (\cos \theta_k - \cos \theta_j) \prod_{h=1}^N \sin \theta_h. \quad (30)$$

Show that (30) can be written as the determinant of a matrix in which θ_j appears only in the j th row, for $j = 1, \dots, N$. You may need the following fact:

$$\det \begin{pmatrix} m_{11} & m_{12} & \cdots & m_{1n} \\ m_{21} & m_{22} & \cdots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ am_{j1} & am_{j2} & \cdots & am_{jn} \\ \vdots & \vdots & \ddots & \vdots \\ m_{n1} & m_{n2} & \cdots & m_{nn} \end{pmatrix} = a \det \begin{pmatrix} m_{11} & m_{12} & \cdots & m_{1n} \\ m_{21} & m_{22} & \cdots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{j1} & m_{j2} & \cdots & m_{jn} \\ \vdots & \vdots & \ddots & \vdots \\ m_{n1} & m_{n2} & \cdots & m_{nn} \end{pmatrix}. \quad (31)$$

Solution: We write

$$\prod_{1 \leq j < k \leq N} (\cos \theta_k - \cos \theta_j) = \det \begin{pmatrix} 1 & \cos \theta_1 & \cdots & \cos^{N-1} \theta_1 \\ 1 & \cos \theta_2 & \cdots & \cos^{N-1} \theta_2 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \cos \theta_N & \cdots & \cos^{N-1} \theta_N \end{pmatrix}. \quad (32)$$

We incorporate the sine functions by multiplying $\sin \theta_j$ through the j th row:

$$\prod_{1 \leq j < k \leq N} (\cos \theta_k - \cos \theta_j) \prod_{h=1}^N \sin \theta_h = \det \begin{pmatrix} \sin \theta_1 & \cos \theta_1 \sin \theta_1 & \cdots & \cos^{N-1} \theta_1 \sin \theta_1 \\ \sin \theta_2 & \cos \theta_2 \sin \theta_2 & \cdots & \cos^{N-1} \theta_2 \sin \theta_2 \\ \vdots & \vdots & \ddots & \vdots \\ \sin \theta_N & \cos \theta_N \sin \theta_N & \cdots & \cos^{N-1} \theta_N \sin \theta_N \end{pmatrix}. \quad (33)$$

The jpdf (ignoring the normalisation constant) is the square of this determinant.

12.7 Gaudin's lemma

To calculate the n -point correlation functions as determinants we will need:

Lemma (Gaudin's lemma). Suppose that we have a function of two variables $f(x, y)$ which is integrable on an interval J and such that

$$\int_J f(x, y) f(y, z) dy = K f(x, z),$$

for all x and z and where $K = K(f, J)$ is a constant. Suppose also that

$$\int_J f(y, y) dy = D,$$

where $D = D(f, J)$ is a constant as well. Then we have

$$\int_J \det_{n \times n} (f(x_j, x_k)) dx_n = (D - (n-1)K) \det_{(n-1) \times (n-1)} (f(x_j, x_k)).$$

Gaudin's lemma is the main ingredient to prove

$$R_n^{U(N)}(\theta_1, \dots, \theta_n) = \det_{n \times n} (S_N(\theta_k - \theta_j)).$$

Indeed, we have

$$\int_0^{2\pi} S_N(\theta_k - \theta) S_N(\theta - \theta_j) d\theta = S_N(\theta_k - \theta_j) \quad (34a)$$

$$\int_0^{2\pi} S_N(\theta - \theta) d\theta = S_N(0) \int_0^{2\pi} d\theta = N. \quad (34b)$$

Eq. (34b) follows from

$$S_N(0) = \lim_{\theta \rightarrow 0} \frac{1}{2\pi} \frac{\sin(\frac{N\theta}{2})}{\sin(\frac{\theta}{2})} = \frac{N}{2\pi}.$$

For Eq. (34a) use

$$S_N(\theta_k - \theta_j) = \frac{e^{i\frac{N-1}{2}(\theta_j - \theta_k)}}{2\pi} \sum_{l=0}^{N-1} e^{il(\theta_k - \theta_j)}.$$

Therefore,

$$\begin{aligned} \int_0^{2\pi} S_N(\theta_k - \theta) S_N(\theta - \theta_j) d\theta &= \frac{e^{i\frac{N-1}{2}(\theta_j - \theta_k)}}{(2\pi)^2} \\ &\times \sum_{l,m=0}^{N-1} e^{i(l\theta_k - m\theta_j)} \int_0^{2\pi} e^{i(m-l)\theta} d\theta = \frac{e^{i\frac{N-1}{2}(\theta_j - \theta_k)}}{2\pi} \sum_{l=0}^{N-1} e^{il(\theta_k - \theta_j)} \\ &= S_N(\theta_k - \theta_j). \end{aligned}$$

Apply Gaudin's lemma to:

$$P_{N2}(\theta_1, \dots, \theta_N) = C(2\pi)^N \det_{N \times N} (S_N(\theta_k - \theta_j)),$$

where $f(x_j, x_k)$ is replaced by $S_N(\theta_k - \theta_j)$, J by $[0, 2\pi)$, $K = 1$ and $D = N$. We have

$$\begin{aligned} \int_0^{2\pi} P_{N2}(\theta_1, \dots, \theta_N) d\theta_N &= C(2\pi)^N \int_0^{2\pi} \det_{N \times N} (S_N(\theta_k - \theta_j)) d\theta_N \\ &= C(2\pi)^N (N - (N - 1)) \det_{(N-1) \times (N-1)} (S_N(\theta_k - \theta_j)) \\ &= C(2\pi)^N \det_{(N-1) \times (N-1)} (S_N(\theta_k - \theta_j)). \end{aligned}$$

Also:

$$\begin{aligned} \int_0^{2\pi} \int_0^{2\pi} P_{N2}(\theta_1, \dots, \theta_N) d\theta_{N-1} d\theta_N &= C(2\pi)^N \int_0^{2\pi} \det_{(N-1) \times (N-1)} (S_N(\theta_k - \theta_j)) d\theta_{N-1} \\ &= C(2\pi)^N (N - (N - 2)) \det_{(N-2) \times (N-2)} (S_N(\theta_k - \theta_j)) \\ &= C(2\pi)^N 1 \cdot 2 \det_{(N-2) \times (N-2)} (S_N(\theta_k - \theta_j)) \end{aligned}$$

By iteration we obtain

$$\begin{aligned} \int_0^{2\pi} \cdots \int_0^{2\pi} P_{N2}(\theta_1, \dots, \theta_N) d\theta_{n+1} \cdots d\theta_N \\ = C(2\pi)^N (N-n)! \det_{n \times n} (S_N(\theta_k - \theta_j)). \end{aligned} \quad (35)$$

By setting $n = 0$ we compute the CUE normalization constant:

$$\int_0^{2\pi} \cdots \int_0^{2\pi} P_{N2}(\theta_1, \dots, \theta_N) d\theta_1 \cdots d\theta_N = C(2\pi)^N N!,$$

therefore $C = 1/((2\pi)^N N!)$. Finally, we have

$$\int_0^{2\pi} \cdots \int_0^{2\pi} P_{N2}(\theta_1, \dots, \theta_N) d\theta_{n+1} \cdots d\theta_N = \frac{(N-n)!}{N!} \det_{n \times n} (S_N(\theta_k - \theta_j)),$$

which gives

$$R_n^{U(N)}(\theta_1, \dots, \theta_n) = \det_{n \times n} (S_N(\theta_k - \theta_j)).$$

13 Averages over $U(N)$ (CUE)

Let $f : U(N) \rightarrow \mathbb{C}$ be a class function, i.e.

$$f(V^\dagger U V) = f(U) \quad V \in U(N). \quad (36)$$

Eq. (36) implies that $f(U) = f(\theta_1, \dots, \theta_N) = f(\theta_{\sigma 1}, \dots, \theta_{\sigma N})$. Essentially these are functions of the eigenvalues of the matrix.

The average of $f(U)$ over $U(N)$ is

$$\int_{U(N)} f(U) d\mu_H(U),$$

where $d\mu_H(U)$ denotes Haar measure on $U(N)$. For class functions the average over the CUE

$$\frac{1}{(2\pi)^N N!} \int_{[0, 2\pi)^N} f(\theta_1, \dots, \theta_N) \prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2 d\theta_1 \cdots d\theta_N.$$

In general, for $p(x_1, x_2, \dots, x_N)$ a *j.p.d.f.*, $x_j \in I$, the average of $f(x_1, x_2, \dots, x_N)$ with respect to that *j.p.d.f.* is

$$\int_{I^N} f(x_1, \dots, x_N) p(x_1, \dots, x_N) dx_1 \cdots dx_N.$$

13.1 Andréief's identity

Lemma (Andréief's identity). *For any interval J and integrable functions $\psi_j(x)$ and $\phi_j(x)$, $1 \leq j \leq n$, we have*

$$\begin{aligned} \frac{1}{n!} \int_{J^n} \det_{n \times n} (\psi_j(x_k)) \det_{n \times n} (\phi_j(x_k)) dx_1 \cdots dx_n \\ = \det_{n \times n} \left(\int_J \psi_j(x) \phi_k(x) dx \right). \end{aligned}$$

The proof is informative but not examinable. It follows below, and you might want to look at the method as it will help you with other questions, but you will not be asked to reproduce it.

Proof (NOT examinable) We have

$$\begin{aligned}
& \int_{J^n} \det_{n \times n}(\psi_j(x_k)) \det_{n \times n}(\phi_j(x_k)) dx_1 \cdots dx_n \\
&= \int_{J^n} \left(\sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n \psi_j(x_{\sigma j}) \right) \left(\sum_{\tau \in S_n} \text{sgn}(\tau) \prod_{k=1}^n \phi_k(x_{\tau k}) \right) \prod_{l=1}^n dx_l \\
&= \int_{J^n} \sum_{\sigma, \tau \in S_n} \text{sgn}(\sigma\tau) \prod_{j=1}^n \psi_j(x_{\sigma j}) \prod_{k=1}^n \phi_k(x_{\tau k}) \prod_{l=1}^n dx_l. \quad (37)
\end{aligned}$$

If G is a group and $g \in G$, then $gG = G$. For example, take S_2 . Its two elements are the identity e and the transposition of two objects τ . Clearly

$$\tau e = \tau \quad \text{and} \quad \tau^2 = e.$$

Therefore $\tau S_2 = S_2$. (Try it with S_3 .)

We now set $\xi = \sigma^{-1}\tau$ or $\tau = \sigma\xi$ in Eq. (37):

$$\int_{J^n} \sum_{\sigma, \xi \in S_n} \text{sgn}(\xi) \prod_{j=1}^n \psi_j(x_{\sigma j}) \prod_{k=1}^n \phi_k(x_{\sigma \xi k}) \prod_{l=1}^n dx_l.$$

Then we substitute $k = \xi^{-1}m$:

$$\begin{aligned}
& \int_{J^n} \sum_{\sigma, \xi \in S_n} \text{sgn}(\xi) \prod_{j=1}^n \psi_j(x_{\sigma j}) \prod_{m=1}^n \phi_{\xi^{-1}m}(x_{\sigma m}) \prod_{l=1}^n dx_l \\
&= \int_{J^n} \sum_{\sigma, \xi \in S_n} \text{sgn}(\xi) \prod_{j=1}^n \psi_j(x_{\sigma j}) \phi_{\xi^{-1}j}(x_{\sigma j}) \prod_{j=1}^n dx_{\sigma j}.
\end{aligned}$$

We use the simple fact that

$$\int_{J^2} f(x)g(y)dx dy = \left(\int_J f(x)dx \right) \left(\int_J g(x)dx \right)$$

and set $\chi = \xi^{-1}$:

$$\begin{aligned}
& \sum_{\sigma, \chi \in S_n} \text{sgn}(\chi) \prod_{j=1}^n \int_J \psi_j(x) \phi_{\chi j}(x) dx \\
&= n! \sum_{\chi \in S_n} \text{sgn}(\chi) \prod_{j=1}^n \int_J \psi_j(x) \phi_{\chi j}(x) dx \\
&= n! \det_{n \times n} \left(\int_J \psi_j(x) \phi_k(x) dx \right).
\end{aligned}$$

This proves Andréief's identity. □

13.2 Averages over $U(N)$ as determinants

We often will encounter integrals that have this form:

$$\frac{1}{n!} \int_{J^n} \prod_{k=1}^n f(x_k) \det_{n \times n}(\psi_j(x_k)) \det_{n \times n}(\phi_j(x_k)) dx_1 \cdots dx_n. \quad (38)$$

We have

$$\prod_{k=1}^n f(x_k) \det_{n \times n}(\psi_j(x_k)) = \det_{n \times n}(f(x_k) \psi_j(x_k)).$$

The integral (38) now becomes

$$\frac{1}{n!} \int_{J^n} \det_{n \times n}(f(x_k) \psi_j(x_k)) \det_{n \times n}(\phi_j(x_k)) dx_1 \cdots dx_n = \det_{n \times n} \left(\int_J f(x) \psi_j(x) \phi_k(x) dx \right) \quad (39)$$

(by Andréief's identity)

Let $f(\theta_1, \dots, \theta_N) = \prod_{j=1}^N f(\theta_j)$. To average this function over $U(N)$ we want to compute

$$\begin{aligned} & \frac{1}{(2\pi)^N N!} \int_{[0, 2\pi)^N} \left(\prod_{j=1}^N f(\theta_j) \right) \prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2 d\theta_1 \cdots d\theta_N \\ &= \frac{1}{(2\pi)^N N!} \int_{[0, 2\pi)^N} \prod_{j=1}^N f(\theta_j) \det_{N \times N} \left(e^{i(j-1)\theta_k} \right) \det_{N \times N} \left(e^{-i(j-1)\theta_k} \right) d\theta_1 \cdots d\theta_N. \\ &= \det_{N \times N} \left(\frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{i(j-k)\theta} d\theta \right) \end{aligned} \quad (40)$$

(applying (39) with $\psi_j(\theta_k) = e^{i(j-1)\theta_k}$ and $\phi_j(\theta_k) = e^{-i(j-1)\theta_k}$) (40) is sometimes called **Heine's identity**

Exercise 8. Compute the average

$$\left\langle \left(\prod_{j=1}^N \cos \theta_j \right) \right\rangle_{U(N)} = \frac{1}{(2\pi)^N N!} \int_{[0, 2\pi)^N} \left(\prod_{j=1}^N \cos(\theta_j) \right) \prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2 d\theta_1 \cdots d\theta_N. \quad (41)$$

You will need the fact that determinants of symmetric tridiagonal matrices obey the recurrence relation

$$x_n = ax_{n-1} - b^2 x_{n-2}, \quad (42)$$

where x_n is the determinant of the matrix of dimension n , a denotes the value of the diagonal elements and b the off-diagonal elements.

Solution: Heine's identity applied to this integral gives

$$\begin{aligned} \left\langle \prod_{j=1}^N \cos(\theta_j) \right\rangle_{U(N)} &= \frac{1}{(2\pi)^N N!} \int_{[0, 2\pi)^N} \left(\prod_{j=1}^N \cos(\theta_j) \right) \prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2 d\theta_1 \cdots d\theta_N \\ &= \det_{N \times N} \left(\frac{1}{2\pi} \int_0^{2\pi} \cos(\theta) e^{i(j-k)\theta} d\theta \right). \end{aligned} \quad (43)$$

To evaluate the matrix elements, write cosine in terms of exponentials and note that most of the integrals are zero. In fact, we end up with a tri-diagonal matrix and everything except the two off-diagonals is zero:

$$\begin{pmatrix} 0 & \frac{1}{2} & & & & \\ \frac{1}{2} & 0 & \frac{1}{2} & & & \\ & \ddots & \ddots & \ddots & & \\ & & \ddots & \ddots & \ddots & \\ & & & \frac{1}{2} & 0 & \frac{1}{2} \\ & & & & \frac{1}{2} & 0 \end{pmatrix}. \quad (44)$$

In our case $a = 0$ and $b = 1/2$, so we simply have

$$x_n = -\frac{1}{4}x_{n-2}, \quad (45)$$

with initial conditions $x_1 = 0$ (because the 1×1 determinant is just a single zero element) and $x_2 = -1/4$ (the determinant of the 2×2 matrix). Therefore, the value of the average (43) is

$$\left\langle \prod_{j=1}^N \cos \theta_j \right\rangle_{U(N)} = \begin{cases} 0 & \text{if } N \text{ is odd} \\ (-\frac{1}{4})^m & \text{if } N = 2m \text{ with } m \geq 1. \end{cases} \quad (46)$$

14 Orthogonal polynomial techniques

We now will compute the correlation functions for any set of N points on an interval $J \subseteq \mathbb{R}$ whose *j.p.d.f.* is of the form

$$P(\lambda_1, \dots, \lambda_N) = C \prod_{j=1}^N w(\lambda_j) \prod_{1 \leq j < k \leq N} |\lambda_k - \lambda_j|^2, \quad (47)$$

where $w(\lambda) \geq 0$ is a real function such that the integral

$$\int_J w(\lambda) \lambda^k d\lambda$$

is finite for any k . $w(\lambda)$ is called *weighting function*.

14.1 Introducing orthogonal polynomials

We have the following theorem.

Theorem. *Given an arbitrary weighting function $w(\lambda)$ there exists a unique (up to a constant of proportionality) sequence of polynomials $p_k(\lambda)$ such that*

$$(p_j, p_k) = \int_J w(\lambda) p_j(\lambda) p_k(\lambda) d\lambda = N_k \delta_{jk}$$

The sequence $p_j(\lambda)$ are called *orthogonal polynomials* with weight $w(\lambda)$. $p_j(\lambda)$ is a polynomial of order j . We will fix the arbitrary constant of proportionality by setting the coefficient of λ^k in $p_k(\lambda)$ equal to one. That is we consider *monic polynomials*.

We denote by $(p_{j-1}(\lambda_k))$ the matrix

$$\begin{pmatrix} p_0(\lambda_1) & p_0(\lambda_2) & \cdots & p_0(\lambda_n) \\ p_1(\lambda_1) & p_1(\lambda_2) & \cdots & p_1(\lambda_n) \\ p_2(\lambda_1) & p_2(\lambda_2) & \cdots & p_2(\lambda_n) \\ \dots & \dots & \dots & \dots \\ p_{n-1}(\lambda_1) & p_{n-1}(\lambda_2) & \cdots & p_{n-1}(\lambda_n) \end{pmatrix},$$

where $p_j(\lambda)$ is a polynomial of degree j . Note that we have already seen matrices of this form: the choice $p_j(\lambda) = \lambda^j$ gives the transpose of the Vandermonde matrix:

$$V^t = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 & \cdots & \lambda_n^2 \\ \dots & \dots & \dots & \dots & \dots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \lambda_3^{n-1} & \cdots & \lambda_n^{n-1} \end{pmatrix}.$$

Theorem. If $\{p_j(\lambda)\}_{j=0,\dots,n-1}$ is a sequence of monic polynomials we have

$$\det_{n \times n}(p_{j-1}(\lambda_k)) = \det_{n \times n}(\lambda_k^{j-1}) = \prod_{1 \leq j < k \leq N} (\lambda_k - \lambda_j).$$

Proof. Let

$$p_j(\lambda) = \lambda^j + a_{j-1}\lambda^{j-1} + a_{j-2}\lambda^{j-2} + \cdots + a_0,$$

where the coefficients a_k s may be different for each polynomial $p_j(\lambda)$. The determinant of a matrix does not change if we add to a row a linear combination of the other rows.

We have that $p_0(\lambda) = 1$ because the $p_j(\lambda)$ s are monic. Thus the first line of (λ_k^j) is left untouched. If $p_{n-1}(\lambda) = \lambda^{n-1} + \sum_{j=0}^{n-2} a_j \lambda^j$ we multiply the first row by a_0 , the second one by a_1 , the third one by a_2 and so on. We then add them to the last row. The n -th row is now $(p_{n-1}(\lambda_1), p_{n-1}(\lambda_2), \dots, p_{n-1}(\lambda_n))$.

We repeat the process for the $(n-1)$ -th row and so on. $\square$

14.2 The joint probability density function as a determinant

Theorem. Let $p_j(\lambda)$ be a set of orthogonal monic polynomials with weight $w(\lambda)$. The following identity holds

$$\prod_{l=1}^N w(\lambda_l) \prod_{1 \leq j < k \leq N} |\lambda_k - \lambda_j|^2 = \left(\prod_{l=0}^{N-1} (p_l, p_l) \right) \det_{N \times N}(K_N(\lambda_j, \lambda_k)), \quad (48)$$

where

$$K_N(\lambda, \mu) = (w(\lambda)w(\mu))^{1/2} \sum_{j=0}^{N-1} \frac{p_j(\lambda)p_j(\mu)}{(p_j, p_j)}. \quad (49)$$

Proof:

Recall that

$$\det_{n \times n}(p_{j-1}(\lambda_k)) = \det_{n \times n}(\lambda_k^{j-1}) = \prod_{1 \leq j < k \leq N} (\lambda_k - \lambda_j).$$

Apply the transposing lemma:

$$\det_{N \times N}(\psi_j(\lambda_k)) \det_{N \times N}(\phi_j(\lambda_k)) = \det_{N \times N} \left(\sum_{l=1}^N \psi_l(\lambda_j) \phi_l(\lambda_k) \right).$$

We have

$$\begin{aligned} \prod_{l=1}^N w(\lambda_l) \prod_{1 \leq j < k \leq N} |\lambda_k - \lambda_j|^2 &= \prod_{l=1}^N w(\lambda_l) \det_{N \times N}(p_{j-1}(\lambda_k)) \det_{N \times N}(p_{j-1}(\lambda_k)) \\ &= \prod_{l=0}^{N-1} (p_l, p_l) \det_{N \times N} \left(\left(\frac{w(\lambda_k)}{(p_{j-1}, p_{j-1})} \right)^{1/2} p_{j-1}(\lambda_k) \right) \\ &\quad \times \det_{N \times N} \left(\left(\frac{w(\lambda_k)}{(p_{j-1}, p_{j-1})} \right)^{1/2} p_{j-1}(\lambda_k) \right). \end{aligned}$$

By applying the transposing lemma, the r.h.s. of the above equation becomes

$$\begin{aligned} \prod_{l=0}^{N-1} (p_l, p_l) \det_{N \times N} \left((w(\lambda_j)w(\lambda_k))^{1/2} \sum_{l=0}^{N-1} \frac{p_l(\lambda_j)p_l(\lambda_k)}{(p_l, p_l)} \right) \\ = \prod_{l=0}^{N-1} (p_l, p_l) \det_{N \times N} (K_N(\lambda_j, \lambda_k)). \quad \square \end{aligned}$$

14.3 Correlation functions

The next important fact we will demonstrate is that the kernel $K_N(\lambda, \mu)$ satisfies the hypotheses of Gaudin's lemma.

We have

$$\begin{aligned} \int_J K_N(\lambda, \mu) K_N(\mu, \nu) d\mu \\ &= (w(\lambda)w(\nu))^{1/2} \\ &\quad \times \sum_{j,k=0}^{N-1} \frac{p_j(\lambda)p_k(\nu)}{(p_j, p_j)(p_k, p_k)} \int_J w(\mu)p_j(\mu)p_k(\mu) d\mu \\ &= (w(\lambda)w(\nu))^{1/2} \sum_{j,k=0}^{N-1} \frac{p_j(\lambda)p_k(\nu)}{(p_j, p_j)(p_k, p_k)} (p_j, p_j) \delta_{jk} \\ &= (w(\lambda)w(\nu))^{1/2} \sum_{j=0}^{N-1} \frac{p_j(\lambda)p_j(\nu)}{(p_j, p_j)} = K_N(\lambda, \nu). \end{aligned}$$

Furthermore,

$$\int_J K_N(\lambda, \lambda) d\lambda = \sum_{j=0}^{N-1} \frac{1}{(p_j, p_j)} \int_J w(\lambda)p_j(\lambda)p_j(\lambda) d\lambda = \sum_{j=0}^{N-1} 1 = N.$$

Assume we have a *j.p.d.f.* $P(\lambda_1, \dots, \lambda_N)$ of the form a normalisation constant multiplying (48). By repeatedly applying Gaudin's lemma, we obtain

$$\begin{aligned} & \int_J \cdots \int_J P(\lambda_1, \dots, \lambda_N) d\lambda_{n+1} \cdots d\lambda_N \\ &= C \left(\prod_{l=0}^{N-1} (p_l, p_l) \right) \int_J \cdots \int_J \det_{N \times N} (K_N(\lambda_j, \lambda_k)) d\lambda_{n+1} \cdots d\lambda_N \\ &= C \left(\prod_{l=0}^{N-1} (p_l, p_l) \right) (N-n)! \det_{n \times n} (K_N(\lambda_j, \lambda_k)). \end{aligned} \quad (50)$$

Integrating over all the N variables, i.e. $n = 0$, we obtain the normalization constant:

$$\int_J \cdots \int_J P(\lambda_1, \dots, \lambda_N) d\lambda_1 \cdots d\lambda_N = C \left(\prod_{j=0}^{N-1} (p_j, p_j) \right) N! = 1,$$

which implies

$$C = \left(\left(\prod_{j=0}^{N-1} (p_j, p_j) \right) N! \right)^{-1}. \quad (51)$$

From the definition of spectral correlation function we have

$$R_n(\lambda_1, \dots, \lambda_n) = \frac{N!}{(N-n)!} \int_J \cdots \int_J P(\lambda_1, \dots, \lambda_N) d\lambda_{n+1} \cdots d\lambda_N$$

and combining Eq. (50) and (51) we obtain

$$R_n(\lambda_1, \dots, \lambda_n) = \det_{n \times n} (K_N(\lambda_j, \lambda_k)). \quad (52)$$

This is not so dissimilar to the previous calculations with the computation of the CUE correlation functions. This is not a coincidence, as we can put those in the setting of orthogonal polynomials on a circle.

Let $w(\theta)$ be a positive periodic function, integrable in $[0, 2\pi)$. We call it weighting function on the circle.

A polynomial of degree k on the unit circle is defined by

$$\phi_k(z) = a_k z^k + a_{k-1} z^{k-1} + \cdots + a_0,$$

where $z = e^{i\theta}$. It turns out that we can define orthogonal polynomials on the circle in an analogous way as on the line.

We have the following theorem:

Theorem. *Given an arbitrary weighting function $w(\theta)$ there exists a unique sequence of polynomials on the circle $\{\phi_k(\lambda)\}$ with the following properties:*

1. *the coefficient of z^k in $\phi_k(z)$ is real and positive;*
2. *the system $\{\phi_k(z)\}$ is orthonormal, i.e.*

$$(\phi_j, \phi_k) = \frac{1}{2\pi} \int_0^{2\pi} w(\theta) \phi_j(\theta) \overline{\phi_k(\theta)} d\theta = \delta_{jk}.$$

The techniques that led to formula (52) are identical if we replace the interval J with the unit circle $\mathbb{R}/\mathbb{Z}$.

The orthogonal polynomials for the CUE have $w(\theta) = 1$.

14.4 Classical $\beta = 2$ ensembles

There are various ensembles that have a *j.p.d.f.* of the form in (47). They are called the $\beta = 2$ ensembles because of the power of 2 on the Vandermonde factor. These are some classical weight functions

$$w(\lambda) = \begin{cases} e^{-\lambda^2} & (-\infty < \lambda < \infty) & \text{Hermite} \\ \lambda^a e^{-\lambda} & (\lambda > 0) & \text{Laguerre} \\ (1-\lambda)^a (1+\lambda)^b & (-1 < \lambda < 1) & \text{Jacobi} \\ (1+\lambda^2)^{-a} & & \text{Cauchy} \end{cases}$$

Each of these weight functions generates a family $\{p_j(\lambda)\}$ of classical orthogonal polynomials and there are matrix ensembles whose spectral correlation functions can be computed using these polynomials.

1. *Hermite polynomials* $\longrightarrow$ GUE ensemble of complex Hermitian matrices. Application to quantum chaos.
2. *Laguerre polynomials* $\longrightarrow$ Laguerre ensemble. They describe the *j.p.d.f.* of an ensemble of *Wishart* matrices, i.e. matrices of the type

$$A = XX^\dagger,$$

where the elements of X are complex Gaussian random variables. With applications to statistics (study of covariance matrices) and engineering.

3. *Jacobi polynomials* $\longrightarrow$ *j.p.d.f.* of the classical compact groups $O^\pm(N)$ and $Sp(2N)$. With applications to number theory, statistical mechanics and mesoscopic systems.
4. *Cauchy polynomials* $\longrightarrow$ certain types of circular ensembles.

The local correlations of these ensembles are not always described by the sine kernel $S(r)$. It depends on the local behaviour of the eigenvalue density.